

GATA6 regulates WNT and BMP programs to pattern precardiac mesoderm during the earliest stages of human cardiogenesis

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eLife Assessment

This **important** study investigates the function of a critical regulator of human early cardiac development. The **convincing** examination of GATA6 function is thorough and well-executed. The study will be of interest to scientists working on how the human heart acquires its identity.

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Abstract

Haploinsufficiency for *GATA6* is associated with congenital heart disease (CHD) with variable comorbidity of pancreatic or diaphragm defects, although the etiology of disease is not well understood. Here, we used cardiac directed differentiation from human embryonic stem cells (hESCs) as a platform to study GATA6 function during early cardiogenesis. GATA6 loss-of-function hESCs had a profound impairment in cardiac progenitor cell (CPC) specification and cardiomyocyte (CM) generation due to early defects during the mesendoderm and lateral mesoderm patterning stages. Profiling by RNA-seq and CUT&RUN identified genes of the WNT and BMP programs regulated by GATA6 during early mesoderm patterning. Furthermore, interactome analysis detected GATA6 binding with developmental transcription factors and chromatin remodelers suggesting cooperative regulation of cardiac lineage gene accessibility. We show that modulating WNT and BMP inputs during the first 48 hours of cardiac differentiation is sufficient to partially rescue CPC and CM defects in *GATA6* heterozygous and homozygous mutant hESCs. This study provides evidence of the regulatory functions for GATA6 directing human precardiac mesoderm patterning during the earliest stages of cardiogenesis to further our understanding of haploinsufficiency causing CHD and the co-occurrence of cardiac and other organ defects caused by human *GATA6* mutations.

Introduction

Congenital heart disease (CHD) is the most common human birth defect, affecting nearly 1% of live births¹. CHD is generally understood to derive from dysfunction in the growth, morphogenesis, and/or differentiation of cardiac-fated progenitor cell derivatives; however, the genetic and molecular networks governing these processes remain under investigation. Treatment options are limited to surgical interventions that can save the life of the infant but do not correct the underlying genetics. The causative genetic variants may contribute to complications and morbidity in adulthood². Research aimed to expand our understanding of the molecular genetics regulating normal human cardiogenesis is thus needed to advance therapeutic strategies in the detection and treatment of CHD.

Studies of cardiac development in model organisms have led to the discovery of key molecular drivers of heart organogenesis, including the GATA family of zinc-finger transcription factors. GATA factors, named by their ability to bind to the consensus sequence (A/T)GATA(A/G)³, represent six family members with GATA4/5/6 expressed in the developing heart of early stage embryos^{4,5}. GATA4/5/6 transcription factors regulate multiple stages of heart development including the specification, proliferation, and differentiation of cardiac progenitor cells (CPCs)⁶. Notably, *GATA6* knockout (KO) mice exhibit early embryonic arrest during primitive streak formation due to defects in visceral endoderm, a phenotype that has complicated the study of the early developmental function of *GATA6* *in vivo*⁷. Mice with conditional deletion of *GATA6* from early CPCs develop heart abnormalities including atrioventricular canal defects, ventricular septal defects, and partial loss of trabeculae^{8,9}. While these *GATA6* conditional homozygous KO mice die at birth due to heart malformations, few cardiac phenotypes were reported in the *GATA6* heterozygous mutants, except for recent reports of bicuspid aortic valve and heart rhythm abnormalities^{10,11}.

In humans, heterozygous mutations in *GATA6* are associated with various forms of CHD including outflow tract (OFT) defects, septal defects, and Tetralogy of Fallot^{12,13}. *GATA6* is required for the development of multiple mesoderm and endoderm derived organs and some CHD patients with heterozygous *GATA6* mutations have comorbidities including pancreatic agenesis, neonatal diabetes, or congenital diaphragmatic hernia^{14,15}. The phenotypic diversity of birth defects in patients containing *GATA6* mutations illustrates the complexity of human haploinsufficiency; there is likely a combination of variants in modifying or interacting genes that converge to influence the disease phenotype^{16,17}. Previous studies of the function of *GATA6* in model organisms have not sufficiently modeled the variety of cardiac (and non-cardiac) defects observed in humans and most describe phenotypes only after homozygous inactivation of the gene. This raises an important question; why do CHDs in humans associate with *GATA6* haploinsufficiency? Species-specific functions for *GATA6* may differ between mouse and humans, for example, raising the question of how observations made in model organisms relate to human cardiac development. There is a clear need to investigate the function of *GATA6* in the context of a human system to better understand the basis for haploinsufficiency.

Current *in vitro* cardiac-directed differentiation protocols for human pluripotent stem cells (hPSCs) provide powerful tools for studying the transcriptional regulatory networks relevant to CHD¹⁸. A previous study using human induced pluripotent stem cells (hiPSCs) with CRISPR-Cas9 induced *GATA6* mutations showed that *GATA6* is required at the stages of CPC and cardiomyocyte (CM) differentiation¹⁹. *GATA6* is also required for definitive endoderm (DE) differentiation toward the pancreatic lineage as reported in several studies using hPSCs^{20–23}. However, the function of *GATA6* during early mesoderm patterning of human cardiac differentiation has not been explored, and how *in vitro* cardiac and DE lineage phenotypes caused by *GATA6* loss-of-function relate to one another is not understood.

To determine the function of GATA6 during the earliest stages of cardiac differentiation, we performed *in vitro* CM directed differentiation using GATA6 loss-of-function human embryonic stem cell (hESC) lines and explored regulatory functions for GATA6 during precardiac mesoderm patterning that is ultimately required for CM generation. Transcriptome analysis revealed that GATA6 mutations impaired expression of genes important for lateral and cardiac mesoderm patterning, and by integrating GATA6 CUT&RUN sequencing with RNA-seq we identified GATA6-regulated targets including those of the WNT and BMP networks, key morphogenetic signaling pathways required to induce anterior primitive streak formation^{24,25}. Furthermore, we found that GATA6 interacts with important developmental transcription factors and chromatin remodelers during early mesoderm patterning stages that may promote DNA accessibility required for cardiac lineage commitment. Taken together, these findings uncover multiple functions for GATA6 in regulating very early precardiac mesoderm patterning and the lineage specific gene networks required for human CM generation.

Results

GATA6 loss-of-function impairs cardiac mesoderm lineage differentiation

A monolayer cytokine-based *in vitro* cardiac differentiation protocol for hPSCs was adapted from previously established methods^{26–28} (Figure 1A). GATA6 heterozygous mutant ($GATA6^{+/-}$), homozygous mutant ($GATA6^{-/-}$), and isogenic wildtype (WT) control ($GATA6^{+/+}$) H1-hESC lines were previously generated^{20,29} (Supplemental Figure 1A, B). These lines were differentiated in parallel to determine the impact of GATA6 loss-of-function during human cardiogenesis. Western blotting at days 2 or 5 of cardiac differentiation confirmed the loss of GATA6 protein in $GATA6^{-/-}$ cells and intermediate levels of protein in $GATA6^{+/-}$ cells relative to WT controls (Supplemental Figure 1C). The percentage of cardiac troponin T positive (%cTnT⁺) CMs in individual differentiation assays remained stable starting around day 13 and thus CM differentiation efficiency was analyzed during a window of days 13 to 18 of cardiac differentiation. $GATA6^{-/-}$ cells failed to generate beating CMs and yielded close to 0% cTnT⁺ CMs when examined by flow cytometry (Figure 1B, C). In contrast, $GATA6^{+/-}$ hESCs did generate beating CMs (~25% cTnT⁺) but at a significantly reduced efficiency compared to WT cells (~55% cTnT⁺, Figure 1B, C). $GATA6^{+/-}$ CMs obtained were comparable morphologically to WT CMs with no obvious sarcomere defects (Supplemental Figure 1D). No significant differences in %cTnT⁺ were observed between clones of the same genotype (Figure 1C), therefore data from lines with the same genotypes were combined in all subsequent comparisons.

We also generated an iPSC line derived from a CHD patient presenting with an atrial septal defect and congenital diaphragmatic hernia; this patient was found to have a heterozygous frameshift mutation in GATA6 (c.1071delG)¹⁵. The mutant allele was corrected to WT sequence ($GATA6^{corr/+}$) using CRISPR-Cas9 gene editing (Supplemental Figure 1E, F). The iPSC colonies expressed normal levels of the pluripotency markers NANOG and SOX2 demonstrating efficient reprogramming (Supplemental Figure 1G). We performed cardiac differentiation using the cytokine-based protocol for $GATA6^{corr/+}$ and $GATA6^{1071delG/+}$ iPSCs to determine how the phenotype compared to the $GATA6^{+/+}$ and $GATA6^{+/-}$ H1-hESCs. To date, this is the first patient-derived iPSC line with a heterozygous GATA6 mutation along with an isogenic mutation-corrected control to be evaluated for cardiac directed differentiation. $GATA6^{1071delG/+}$ cells generated cTnT⁺ CMs less efficiently than $GATA6^{corr/+}$ cells, on average about 50% (Supplemental Figure 1H), indicating that the GATA6 c.1071delG mutation similarly disrupts cardiac differentiation as observed in the hESC heterozygous mutant lines. Likewise, the CMs obtained by differentiation of either $GATA6^{1071delG/+}$ or $GATA6^{corr/+}$ iPSCs appeared morphologically normal (Supplemental Figure 1I). The fact that both types of heterozygous cells (patient-derived iPSCs and targeted H1-hESCs) generate similar defects in CM differentiation provides evidence supporting the use of the

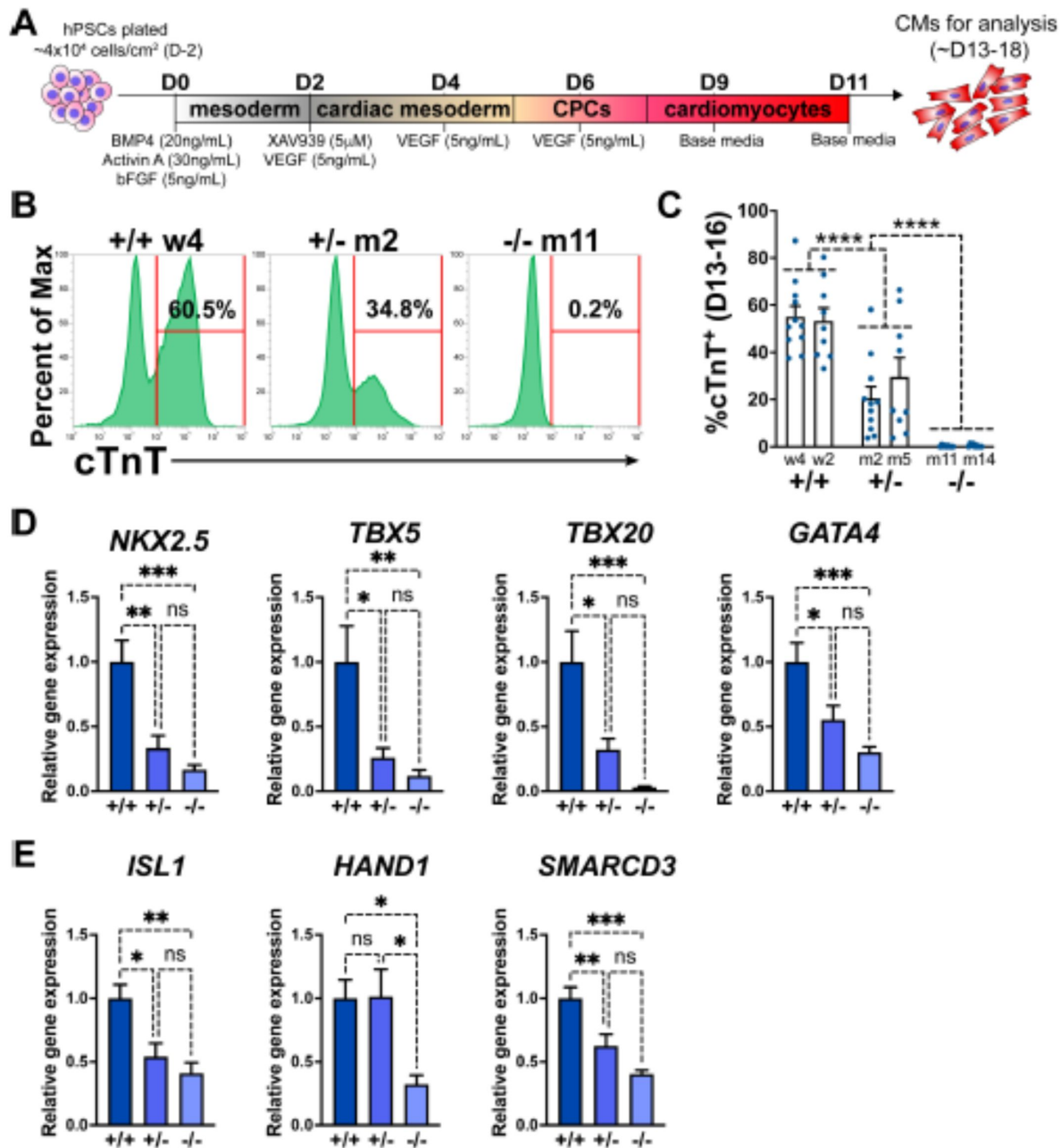


Figure 1.

GATA6 loss-of-function inhibits CM and CPC development.

(A) Schematic for *in vitro* CM directed differentiation cytokine-based protocol (used throughout the study except for Supplemental Fig. 2B-E). Color gradients represent relative developmental stages (white: pluripotent, grey: mesoderm, yellow: cardiac mesoderm, pink: CPCs, and red: CMs). (B) Representative flow cytometry plots for cTnT⁺ CMs at day 14 of cardiac differentiation for *GATA6*^{+/+}, *GATA6*^{+/-}, and *GATA6*^{-/-} hESCs. (C) %cTnT⁺ CMs quantified by flow cytometry between days 13 and 16 (dots indicate independent biological replicates). Significance indicated as *****p*<0.0001 by two-way ANOVA with Tukey's multiple comparison test by genotype. There was no significant difference between clones of the same genotype when two-way ANOVA and Tukey's multiple comparison test was performed for all six sample groups. (D) Day 6 RT-qPCR for the CPC markers indicated normalized to *GATA6*^{+/+} (*n*=6). (E) Day 6 RT-qPCR for heart field markers normalized to *GATA6*^{+/+} (*n*=4). Data represents the mean $\pm$ SEM, significance indicated as **p*<0.05, ***p*<0.01, ****p*<0.001, ns indicates not significant as determined by one-way ANOVA and Tukey's multiple comparisons test. The labels w4, w2 (wildtype), m2, m5 (heterozygous), and m11, m14 (homozygous) refer to the isogenic wildtype and mutant hESC clones (see Supplemental Fig. 1B)

H1-hESCs and isogenic lines to study the genetic and cellular basis for CHD. The corrected iPSC line was shown to have a grossly normal karyotype and to restore higher levels of GATA6 protein as expected (Supplemental Figure 1J, K).

The impaired CM generation from *GATA6* mutant hESCs may be due to deficient CPC specification. Therefore, RT-qPCR assays were performed on *GATA6*^{+/+}, *GATA6*^{+/-} and WT cells harvested at day 6 of cardiac differentiation to examine the expression levels of the CPC markers *NKX2.5*, *TBX5*, *TBX20*, and *GATA4*. CPC marker expression levels were significantly reduced in *GATA6*^{+/+} and *GATA6*^{+/-} cells relative to WT controls, indicating a defect in CPC specification (**Figure 1D**). At this time point there was a clear trend but not a statistically significant difference in the expression levels of CPC markers comparing the *GATA6*^{+/+} and *GATA6*^{+/-} genotypes. We also analyzed expression levels for markers of the first heart field (*HAND1* and *SMARCD3*) and the second heart field (*ISL1*). All were significantly depleted in the *GATA6*^{+/-} cells, while both *ISL1* and *SMARCD3* (but not *HAND1*) levels were significantly lower in the *GATA6*^{+/+} cells. We note that an RT-qPCR time course showed that CPC marker expression level was increased (although not to WT levels) in *GATA6*^{+/+} cells at later stages (days 8 to 11) while *GATA6*^{+/-} CPC gene expression levels remained “flat” in comparison with no increase over time (Supplemental Figure 2A).

GATA6^{+/+}, *GATA6*^{+/-}, and *GATA6*^{-/-} hESCs were differentiated toward the CM lineage using an alternative protocol based on chemical manipulation of the WNT pathway³⁰ (CHIR protocol, Supplemental Figure 2B-E) to confirm if the phenotype is consistent as observed using the cytokine-based protocol. *In vitro* cardiac differentiation protocols for hPSCs are known to sometimes require extensive optimization due to cell line variation and can require testing various cell seeding densities, media component concentrations, and batch variations for crucial reagents^{31,32}. The CHIR protocol was overall suboptimal for cardiac differentiation using the H1-hESCs due to substantial variation in CM differentiation efficiency and the occasional failure to generate CMs in WT H1-hESCs. We therefore committed to using the cytokine-based protocol (**Figure 1A**) for the bulk of experiments in this study because it proved to be a more consistent CM differentiation protocol for the H1-hESCs. Nevertheless, in successful experiments performed using the CHIR protocol we observed the same phenotype in *GATA6*^{-/-} cells and *GATA6*^{+/+} cells using the CHIR protocol as with the cytokine-based protocol (data not shown), consistent with a previous report¹⁹. Bulk RNA-seq was performed at day 5 using the CHIR protocol (Supplemental Figure 2C-E). Principal component analysis (PCA) showed that the cells derived from each genotype were well separated from each other (Supplemental Figure 2C). Differential gene expression and gene ontology (GO) analyses revealed that *GATA6*^{-/-} cells had strongly reduced expression levels of cardiac development related gene sets including “OFT septum morphogenesis” and “heart development” compared to WT controls (Supplemental Figure 2D). Furthermore, while *GATA6*^{+/+} cells had relatively higher expression levels for many of these genes compared to *GATA6*^{-/-} cells, numerous CPC regulatory genes showed decreased levels compared to WT controls (including *NKX2.5*, *TBX20*, *TBX5*, *MEF2C*, and *HAND2*; Supplemental Figure 2E) confirming an impairment of CPC development in differentiating *GATA6* mutant hESCs. GO analysis of genes with increased expression levels in *GATA6*^{-/-} cells relative to WT included gene sets such as “nervous system development” (Supplemental Figure 2D) suggesting a bias towards a non-cardiac genetic signature. Notably, gene sets related to retinoic acid (RA) signaling were significantly increased in *GATA6*^{-/-} cells relative to WT (Supplemental Figure 2D), and *ALDH1A2* expression (the main enzyme that generates RA) was increased in *GATA6*^{-/-} cells relative to WT from days 4 to 6 using the cytokine-based protocol (Supplemental Figure 2F), consistent with a previous study of cardiac differentiation using *GATA6* loss-of-function hiPSCs¹⁹. Proper RA signaling is crucial for cardiogenesis including OFT development and cardiac chamber morphogenesis³³, and inhibiting excessive RA signaling in *GATA6* mutant hiPSCs was shown to partially rescue CPC marker expression¹⁹. Thus, upregulated RA signaling in *GATA6*^{-/-} hESCs during the CPC developmental stage may partially underlie the CM differentiation defects.

Examining protein expression at multiple stages during cardiac differentiation revealed the earliest (and highest) expression level for GATA6 at day 2 of differentiation (**Figure 2A** [↗](#)). The impaired CPC development in *GATA6*^{+/-} and *GATA6*^{-/-} cells may therefore be due to defective cardiac mesoderm patterning during the early stages of cardiogenesis. We examined co-expression of the precardiac mesoderm markers KDR and PDGFRα (KP)²⁷ [↗](#) using flow cytometry from day 3 to 5 of cardiac differentiation (**Figure 2B** [↗](#)). At day 3, there was no significant difference in the %K⁺P⁺ cells between genotypes (~45% K⁺P⁺), but by day 5 the %K⁺P⁺ increased to ~72% in WT and *GATA6*^{+/-} cells, while the %K⁺P⁺ was markedly reduced in *GATA6*^{-/-} cells (~22%, **Figure 2B** [↗](#), C). This is consistent with previous reports where sorting and reseeded K⁺P⁺ cells at day 3 of cardiac differentiation generated both mesenchymal cells and CMs but day 5 sorted K⁺P⁺ cells gave rise predominantly to CMs, indicating that day 5 KP expression is relatively enriched for precardiac mesoderm²⁷ [↗](#),³⁴ [↗](#). *GATA6*^{+/-} cells had a small but significant decrease in the %K⁺P⁺ cells at day 4 relative to WT (Supplemental Figure 3A, B) suggesting that the developing K⁺P⁺ cardiac mesoderm in *GATA6*^{+/-} cells is temporarily defective or delayed relative to WT controls. Notably, the decreased %K⁺P⁺ observed at day 4 (*GATA6*^{+/-} and *GATA6*^{-/-} cells) and day 5 (*GATA6*^{-/-} cells) relative to WT cells was due to reduced KDR but not PDGFRα expression (**Figure 2D** [↗](#), E, Supplemental Figure 3A, B), indicating that KDR is (directly or indirectly) regulated by GATA6 during cardiac mesoderm patterning.

We next sought to determine whether GATA6 loss-of-function impaired mesoderm induction by analyzing the expression of the pan-mesoderm marker BRACHYURY (*T*). Flow cytometry performed on WT, *GATA6*^{+/-}, or *GATA6*^{-/-} cells at day 2 or 3 of cardiac differentiation revealed no significant difference in abundance of BRACHYURY⁺ cells between genotypes (**Figure 2F** [↗](#), G). Likewise, there was no significant change in transcript levels for *T* or the mesendoderm marker *EOMES*, or mesoderm markers *MESP1*, or *MESP2* at day 2 when assessed by RT-qPCR (**Figure 2H** [↗](#)). These results indicate that *GATA6* mutant hESCs have the capacity to differentiate into a nascent mesendoderm/early mesoderm population that is deficient in patterning cardiac mesoderm.

Transcriptome analysis of *GATA6* mutant hESCs during early mesoderm patterning

During gastrulation, sequentially lineage-restrictive mesodermal cell populations emerge along a developmental trajectory involving primitive streak to lateral mesoderm formation, followed by cardiac mesoderm patterning³⁵ [↗](#). As *GATA6* loss-of-function hESCs differentiated to an emergent mesendoderm/mesoderm stage but failed to properly generate cardiac mesoderm, we hypothesized that GATA6 is required during the earliest stages of lateral and precardiac mesoderm patterning. GATA6-regulated genes important during these early mesoderm patterning stages were investigated by performing bulk RNA-seq at day 2 or day 3 of cardiac differentiation (Supplemental Figure 4A, B). Again, PCA showed the profiles representing each genotype were well separated.

GO analysis and gene set enrichment analysis (GSEA) using the biological process (BP) subcollection for differentially expressed genes (DEGs; defined as $p\text{-adj} < 0.05$ and $\log_2(\text{fold change}) > 0.5$) comparing *GATA6*^{-/-} to WT samples revealed decreased enrichment for GO terms including “heart development” (e.g., *HAND1*) and “OFT septum morphogenesis” (e.g., *NRP1*) at days 2 and 3 suggesting a disruption of early cardiac signature genes (**Figure 3A** [↗](#), C, D). The decreased enrichment for the “OFT septum morphogenesis” GO term is consistent with the conclusions of a previous study of *GATA6* mutations in hPSCs in the context of cardiac differentiation¹⁹ [↗](#); however the results presented here implicate this program at a much earlier developmental stage (D2 mesendoderm/lateral mesoderm). Conversely, GO analysis of increased DEGs in *GATA6*^{-/-} samples relative to WT at days 2 and 3 included “extracellular matrix (ECM)

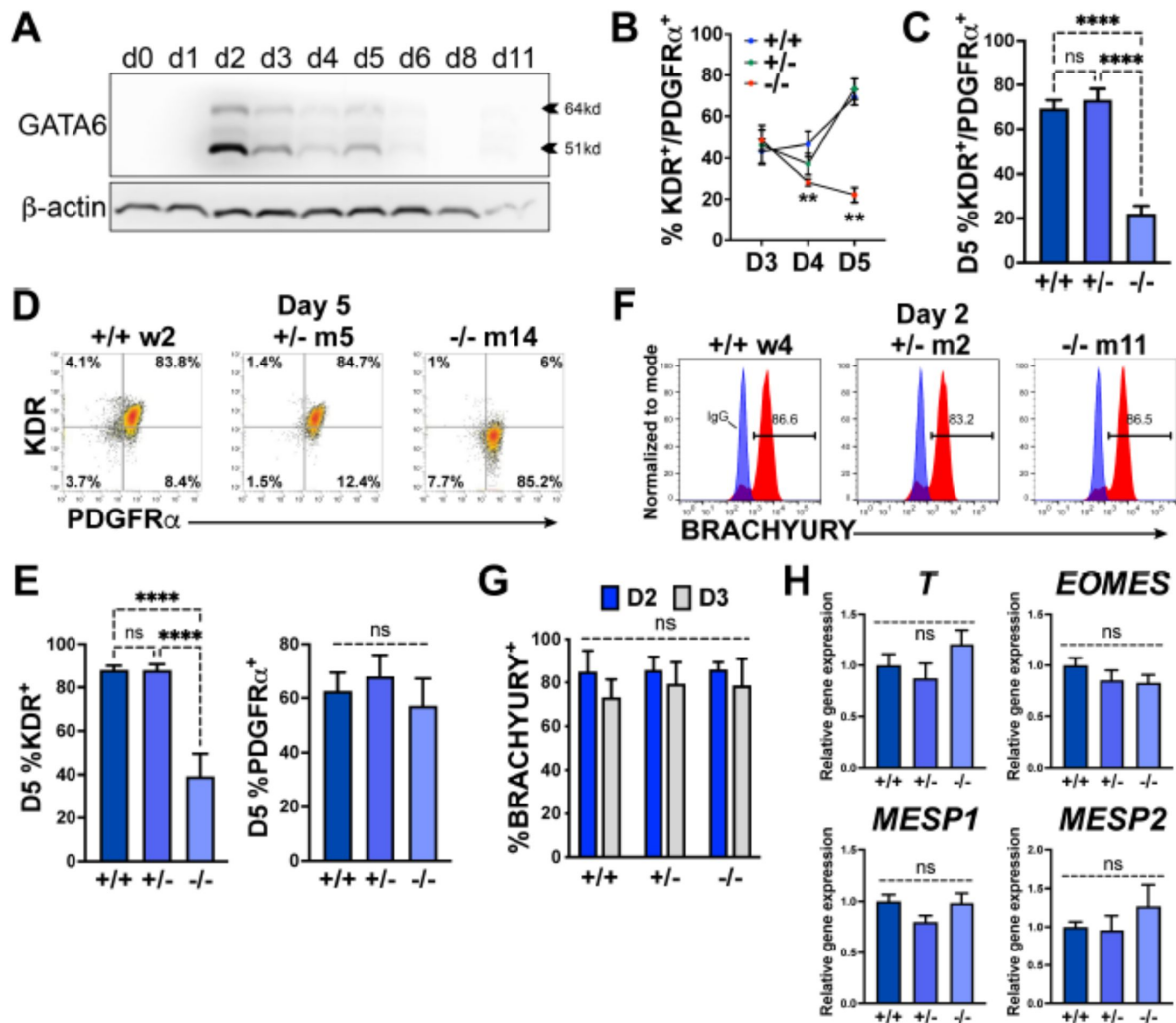


Figure 2.

GATA6 is required for cardiac mesoderm development.

(A) Western blot time course using protein lysates from *GATA6*^{+/+} hESCs probed for GATA6 with β-actin used as a loading control. (B) Flow cytometry quantification for % KDR and PDGFRα double-positive (%K⁺P⁺) cells from days 3 to 5 of cardiac differentiation of *GATA6*^{+/+}, *GATA6*^{+/-}, or *GATA6*^{-/-} hESCs (n≥4). Asterisks indicate statistical significance comparing *GATA6*^{-/-} and WT on days 4 or 5 of cardiac differentiation. (C) Day 5 flow cytometry quantification for %K⁺P⁺ cells (n=7). (D) Representative flow cytometry plots for day 5 %K⁺P⁺ cells. (E) Day 5 flow cytometry quantification (n=7) for %KDR⁺ (left) or PDGFRα⁺ (right). (F) Representative flow cytometry plots for day 2 %BRACHYURY⁺ cells (red) overlaid IgG stained controls (blue). (G) Quantification for day 2 or day 3 %BRACHYURY⁺ cells (n=4). (H) RT-qPCR for day 2 *T*, *EOMES*, *MESP1*, and *MESP2* expression levels normalized to *GATA6*^{+/+} samples (n=6). Data represents the mean ± SEM, with significance indicated as **p<0.01, ****p<0.0001, and ns indicating not significant by two-way ANOVA (B and G) or one-way ANOVA (C, E, H, and I) with Tukey's multiple comparison test. The labels w4, w2 (wildtype), m2, m5 (heterozygous), and m11, m14 (homozygous) refer to the isogenic wildtype and mutant hESC clones (see Supplemental Fig. 1B)

organization”, “skeletal system development”, and “AP axis specification” (**Figure 3B** [↗](#), D), suggesting that *GATA6*^{-/-} cells are biased to other lineages such as paraxial mesoderm or retain an early mesodermal phenotype by day 3.

GATA6^{+/-} samples also had reduced expression levels for genes associated with cardiac GO terms such as “cardiac muscle contraction” and “OFT septum morphogenesis” when comparing to WT controls at day 3 (Supplemental Figure 4C, D). Indeed, there was overlap for many of the decreased DEGs (dDEGs) observed in *GATA6*^{+/-} and *GATA6*^{-/-} cells at day 2 or day 3 (relative to WT) including cardiac development genes such as *ERBB4* (important for endocardial cushion and ventricular trabeculae development³⁶ [↗](#)), the second heart field (SHF) related transcription factor *HAND2*, and the cardiac mesoderm marker *LRRC32*³⁷ [↗](#); although the decrease in expression levels for these genes relative to WT was not as severe in the *GATA6*^{+/-} genotype as for *GATA6*^{-/-} (Supplemental Figure 4D, E, F).

We observed dysregulation of genes related to the WNT and BMP networks in *GATA6*^{-/-} samples relative to WT at day 2 (Supplemental Figure 4G) and GO and GSEA analyses revealed decreased levels of “positive regulation of the BMP pathway” and increased levels of “negative regulation of canonical WNT pathway” at days 2 or 3 relative to WT (**Figure 3A** [↗](#), B, D, E). Several dDEGs identified in *GATA6*^{-/-} cells at day 2 or day 3 relative to WT samples included negative regulators of WNT signaling, including *SFRP1* and *TLE1* (Supplemental Figure 4E, G), and antagonists of the BMP pathway, such as *BMPER* and *SMAD6* (**Figure 3A** [↗](#), E, Supplemental Figure 4E, G), indicating that the overall transcriptional response induced by the WNT and BMP pathways is dysregulated. *GATA6*^{+/-} cells also exhibited perturbed gene expression for several WNT- and BMP-related genes (at an intermediate level to those observed in *GATA6*^{-/-} cells) at days 2 or 3 relative to WT (**Figure 3E** [↗](#), Supplemental Figure 4G). Mesoderm patterning is controlled by the input of WNT, BMP, and TGFβ signaling and dysfunction in these pathways cause severe developmental defects^{38,39} [↗](#). Therefore the cardiac defects caused by *GATA6* loss-of-function may be caused by these alterations in WNT and BMP signaling.

We examined activity of the BMP, TGFβ and WNT pathways by analyzing the effector proteins SMAD1/5/9 (BMP activated), SMAD2/3 (TGFβ activated), and β-catenin (WNT activated) by western blotting at day 2. There was no change in the levels of active phospho-SMAD1/5/9, phospho-SMAD2/3, or non-phospho-β-catenin when comparing *GATA6*^{+/-}, *GATA6*^{-/-} and WT cells (**Figure 3F** [↗](#)). These results indicate that *GATA6* regulates the transcriptional response of genes activated by the BMP and WNT pathways downstream of canonical effector protein activation.

To further examine the impact of *GATA6* loss-of-function on early mesoderm patterning, we correlated RNA-seq data from a study examining hESC differentiation toward cardiac mesoderm⁴⁰ [↗](#) to our day 2 or day 3 RNA-seq data. GSEA analysis using increased DEGs at day 2 (lateral mesoderm) or day 3 (precardiac mesoderm) from Koh et al.⁴⁰ [↗](#) revealed significantly reduced enrichment in our *GATA6*^{-/-} samples (relative to WT) on days 2 or 3, respectively (**Figure 3G** [↗](#), H). Furthermore, *GATA6*^{+/-} cells had partially reduced expression for these lateral and precardiac mesoderm genes relative to WT cells (**Figure 3H** [↗](#)), suggesting that the impaired CPC and CM differentiation observed for *GATA6*^{+/-} hESCs originates as disrupted lateral and cardiac mesoderm patterning. We additionally compared the increased DEGs at day 2 from Koh et al. for paraxial mesoderm and our day 2 RNA-seq data and observed increased expression for these genes in *GATA6*^{-/-} cells relative to controls (**Figure 3I** [↗](#)), further supporting the notion that the *GATA6*^{-/-} hESCs are at biased toward a paraxial mesoderm-like genetic signature.

A previous study of *GATA6* function during DE differentiation from hESCs reported an interaction between *GATA6*, *EOMES*, and *SMAD2/3* to co-regulate genes important during mesendoderm and transition to DE development²² [↗](#). DE and cardiac mesoderm originate from a bipotent germlayer mesendoderm population⁴¹ [↗](#) and likely share a gene expression signature at this early stage. We therefore performed GSEA analysis using the CHIP-seq dataset for triple-overlap of *GATA6*,

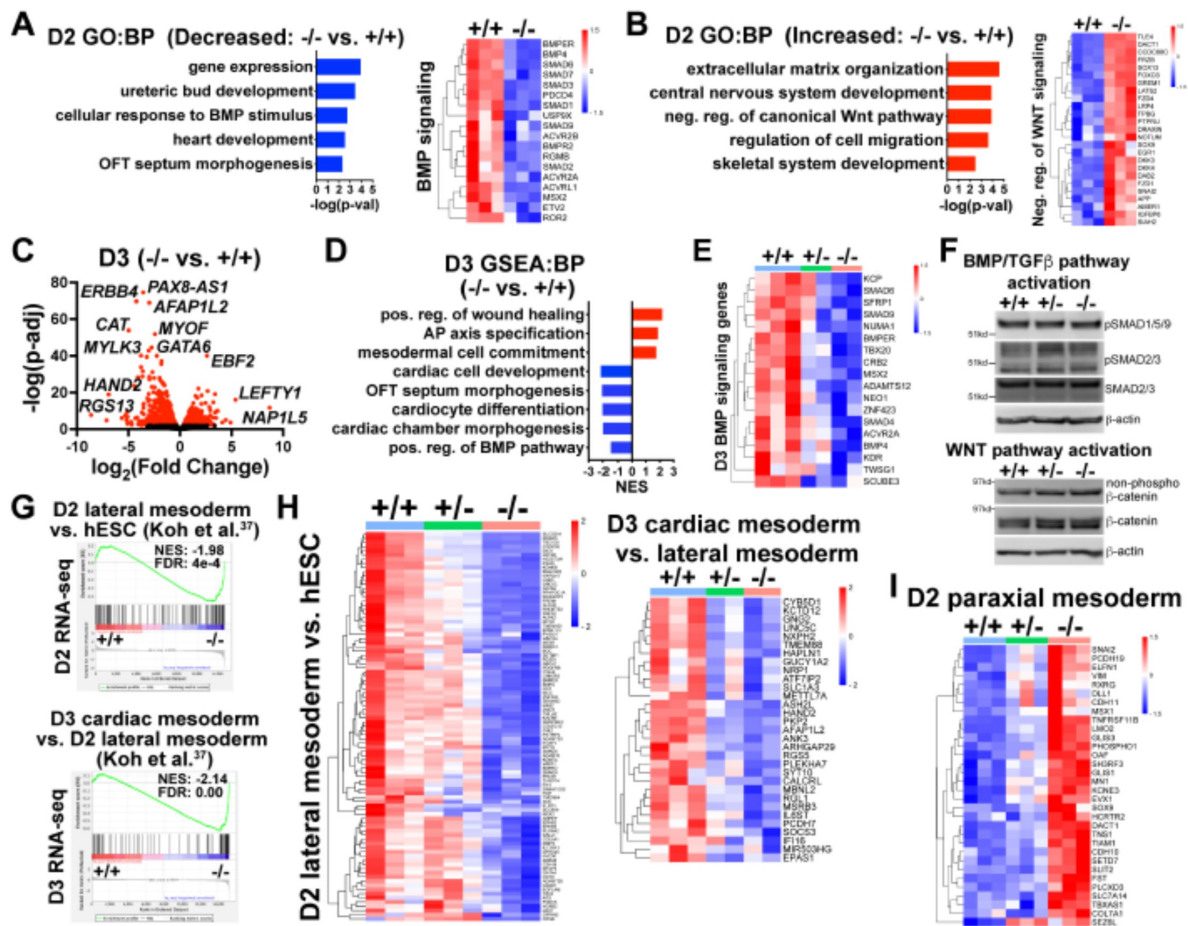


Figure 3.

Transcriptome analysis at early mesoderm patterning stages.

(A) Gene Ontology (GO) analysis for biological process (BP) from decreased differentially expressed genes (dDEGs) identified comparing *GATA6*^{-/-} to WT samples from day 2 RNA-seq data (left). Heatmap for genes related to BMP signaling from day 2 RNA-seq shown on the right. Color gradient on heatmap indicates relative gene expression levels. (B) GO analysis (BP) of increased DEGs identified comparing *GATA6*^{-/-} to WT samples from day 2 RNA-seq (left). Heatmap for genes related to negative regulation of canonical WNT pathway from day 2 RNA-seq shown on the right. (C) Volcano plot for day 2 *GATA6*^{-/-} sample RNA-seq gene expression data relative to WT controls. Dots represent genes, red indicates $p\text{-adj} < 0.05$ and black indicates $p\text{-adj} > 0.05$. (D) Gene set enrichment analysis (GSEA) (BP) of *GATA6*^{-/-} cells relative to WT controls from day 3 RNA-seq data. NES indicates normalized enrichment score. (E) Heatmap for BMP signaling genes from day 3 RNA-seq data. (F) Western blots using protein lysates from *GATA6*^{+/+}, *GATA6*^{+/-} or *GATA6*^{-/-} cells at day 2 of cardiac differentiation probed with antibodies recognizing phospho-SMAD1/5/9, phospho-SMAD2/3, total SMAD2/3, non-phospho- β -catenin, and total β -catenin with β -actin used as a loading control. (G) GSEA analysis of day 2 or day 3 RNA-seq gene expression data (*GATA6*^{-/-} relative to WT) using the day 2 lateral mesoderm (relative to hESC, left) and day 3 cardiac mesoderm (relative to day 2 lateral mesoderm, right) during hESC cardiac differentiation datasets from Koh et al. (Ref. 37) NES indicates normalized enrichment score; FDR indicates false discovery rate. (H) Heatmaps from RNA-seq day 2 (left) or day 3 (right) data using core enrichment genes identified in (G) for lateral mesoderm (relative to hESCs) and cardiac mesoderm (relative to lateral mesoderm). (I) Heatmap showing enriched paraxial mesoderm gene expression in the mutant cells.

EOMES, and SMAD2/3 during early DE differentiation from Chia et al.²² and found a significantly decreased enrichment in *GATA6*^{-/-} cells relative to WT at day 2 for putative direct GATA6 target genes (Supplemental Figure 4H). EOMES is an important transcriptional regulator of anterior mesoderm development⁴², and as GATA6 was shown to interact with EOMES during early DE differentiation^{22,43} we hypothesized that an interaction is similarly necessary during early lateral mesoderm patterning. We therefore performed GSEA using another published dataset of EOMES targets identified during mesoderm and DE (ME) differentiation in mouse ESCs (mESCs)⁴⁴. GSEA analysis of our day 2 RNA-seq data with the “EOMES ME direct activation dataset” human gene equivalents revealed a significant decrease in enrichment in *GATA6*^{-/-} cells relative to WT (Supplemental Figure 4H), supporting the notion that GATA6 cooperates with EOMES to co-regulate genes required for lateral mesoderm patterning.

GATA6 binds to putative regulatory regions associated with BMP, WNT, and EOMES related dDEGs

To determine the genomic localization of GATA6 occupancy during mesoderm patterning, we performed cleavage under targets and release using nuclease (CUT&RUN) sequencing for GATA6 at day 2 of differentiation using *GATA6*^{+/+} hESCs. 1,273 significant differentially bound sites were identified for GATA6 relative to IgG control samples, associated with 953 transcriptional start sites (TSS) for unique genes (Figure 4A). Significant GATA6 binding sites predominately localized to intronic (46.03%) and distal intergenic (32.21%, >5kb from TSS) regions (Figure 4A), consistent with previous reports of GATA6 binding distribution^{45,46}. Transcription factor binding motif enrichment analysis revealed the strongest enrichment at GATA6 bound loci for “GATA” motifs, as well as “LHX-like” (LIM homeobox transcription factor; expressed in cardiac and other mesodermal derivatives⁴⁷) and “EOMES” motifs (Figure 4B). GO analysis of the gene list nearby significant GATA6 binding peaks showed striking similarity to GO analysis from RNA-seq dDEGs in *GATA6*^{-/-} cells (relative to WT), including “OFT septum morphogenesis”, “BMP signaling pathway” and “WNT signaling pathway” (Figure 4C). Additionally, we observed an overlap of CUT&RUN identified genes with day 2 (17.9%) and day 3 (16.8%) RNA-seq dDEGs in *GATA6*^{-/-} cells (relative to WT) representing high likelihood direct targets of GATA6 including the cardiac development related genes *HAND1* and *MYOCD* (Figure 4D). Visualizing the localization of these GATA6 peaks that mapped to day 2 and day 3 dDEGs confirmed binding in intronic regions or at distal intergenic regions that overlapped with known enhancer-like signatures identified by the ENCODE Project⁴⁸ (Figure 4E). These included several WNT-related genes such as *MCC* and *LGR5*, as well as BMP-related genes including *SMAD6* (Figure 4E). In Figure 4F genes are highlighted in red that have GATA6 binding peaks in associated putative enhancers that are likely to be direct targets for the WNT/BMP networks. These are distinguished from WNT/BMP pathway genes that were not bound by GATA6 yet are down-regulated in the *GATA6* mutant cells and are more likely to be indirect targets.

Day 2 GATA6 CUT&RUN data revealed an enrichment for binding to sequences containing EOMES motifs (Figure 4B). We compared our day 2 GATA6 CUT&RUN and RNA-seq data to previously published EOMES CHIP-seq data performed on hESCs at day 2 of DE differentiation⁴⁹. Comparing these datasets revealed 32 genes that had triple-overlap of day 2 GATA6 CUT&RUN identified genes (953 genes nearby significantly bound sites), day 2 RNA-seq dDEGs, and EOMES CHIP-seq targets⁴⁹ (Figure 4G). Among this list of triple-overlap genes are known cardiac development factors including *LGR5*, which was reported to be required for hESC-CM differentiation⁵⁰, and *PRDM1*, which is expressed in the SHF and involved in OFT morphogenesis⁵¹. Visualizing the location of these triple-overlap genes confirmed that the GATA6 peaks directly overlapped with many EOMES CHIP-seq peaks (Supplemental Figure 5), suggesting that GATA6 and EOMES interact to co-regulate genes required for precardiac mesoderm patterning.

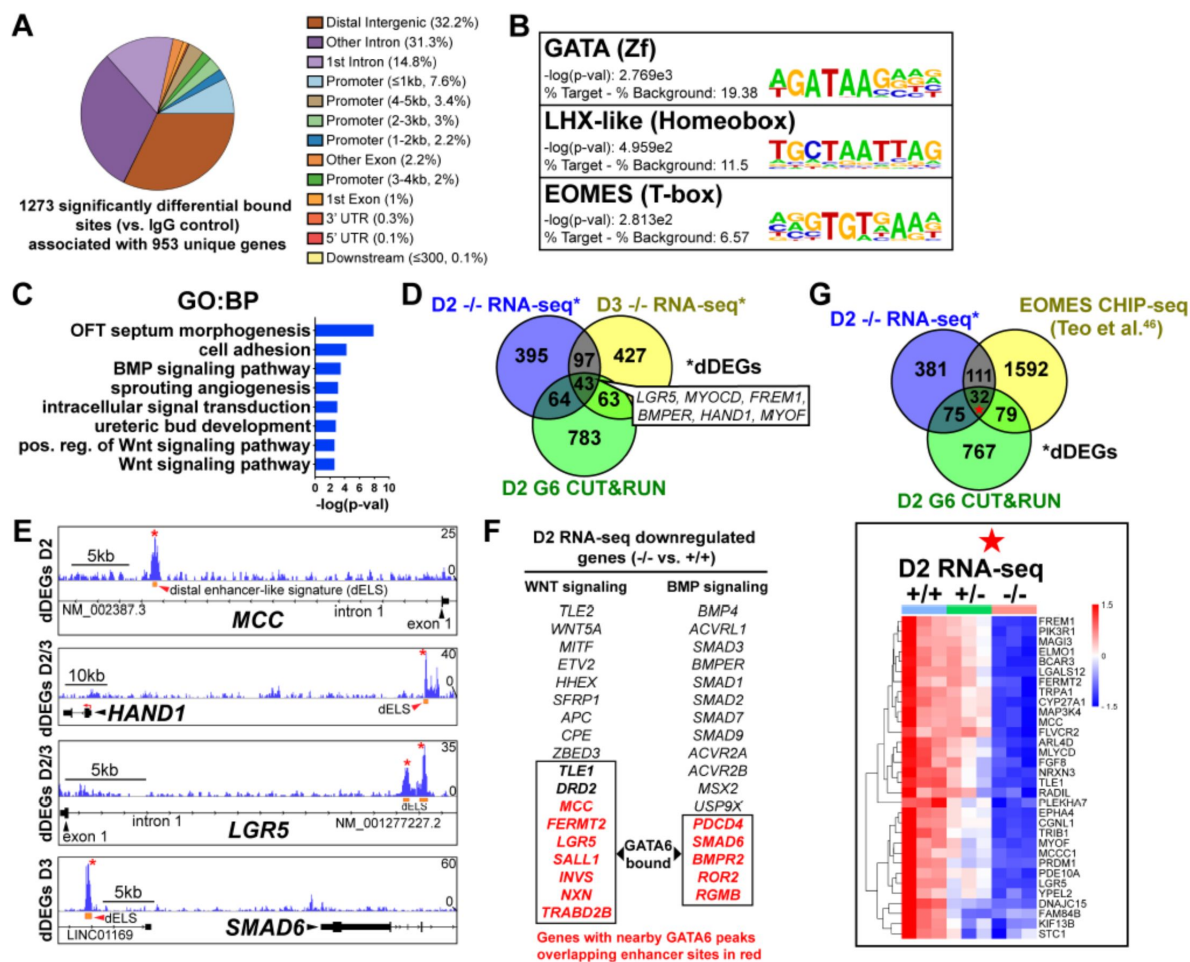


Figure 4

GATA6 CUT&RUN analysis during early mesoderm patterning.

(A) Genomic distribution for significant GATA6 binding peaks identified by GATA6 CUT&RUN at day 2 of cardiac differentiation ($n=3$). (B) Transcription factor motif enrichment at GATA6 bound loci. (C) GO (BP) analysis of the gene list associated with significant GATA6 binding peaks. (D) Venn diagram comparisons for day 2 GATA6 CUT&RUN identified genes (green), day 2 dDEGs (blue) and day 3 dDEGs (yellow) identified by RNA-seq ($GATA6^{-/-}$ relative to WT). (E) Human genome browser representations of GATA6 CUT&RUN data (blue tracks) aligned to select genes that are dDEGs identified by day 2 or 3 RNA-seq ($GATA6^{-/-}$ relative to WT). Orange rectangles represent approximate location for distal enhancer-like signatures (dELS) via the ENCODE Project. Asterisks indicate significant GATA6 binding peaks ($p<0.003$ relative to IgG controls). (F) Genes from the WNT and BMP signaling pathways that are significantly downregulated in the mutant cells compared to wildtype. Genes in Red were found by CUT&RUN to have GATA6 binding peaks in associated putative enhancers and are therefore likely to be direct targets. (G) Venn diagram comparisons for day 2 GATA6 CUT&RUN identified genes (green), day 2 dDEGs (blue), $GATA6^{-/-}$ relative to WT) and EOMES-CHIP-seq identified genes at day 2 of hESC-DE differentiation from Teo et al. (Ref. 46) (yellow). Inlayed heatmap indicates $GATA6^{+/+}$, $GATA6^{+/-}$ and $GATA6^{-/-}$ RNA-seq gene expression data at day 2 of cardiac differentiation for the 32 triple-overlap genes.

GATA6 interactome analysis during precardiac to cardiac mesoderm patterning stages

We next analyzed the GATA6 interactome using rapid immunoprecipitation of endogenous proteins (RIME) on WT cells at day 2 or day 4 of cardiac differentiation to define the chromatin associated protein interactions. 99 significant interacting proteins at day 2 and 52 unique proteins at day 4 of cardiac differentiation were identified bound to GATA6 compared to IgG control samples (**Figure 5A**). EOMES was one of the most enriched proteins bound to GATA6 at day 2 but was not identified in the day 4 GATA6-RIME analysis (**Figure 5B**). 81.8% of proteins identified by GATA6-RIME at day 2 were not encoded by significantly differentially expressed genes identified by day 2 RNA-seq (*GATA6*^{-/-} vs. WT) while 18.2% were, including TLE1, VRTN, and ZIC2 (Supplemental Figure 6A), indicating that GATA6 functions in parallel with most interacting proteins identified. Many of the significantly bound proteins were components of chromatin remodeling complexes such as members of the SWI/SNF complex (e.g., SMARCC1, SMARCA4, and ARID1A), which functions to mobilize nucleosomes and promote DNA accessibility⁵², and the Nucleosome Remodeling and Deacetylase (NuRD) complex (e.g., KDM1A, MTA1/2/3, and GATAD2A), a chromatin remodeling complex associated with transcriptional repression⁵³ (**Figure 5B**). GATA6 protein interactions were also observed for other epigenetic and transcriptional regulators including those related to histone deacetylation (i.e., HDAC2) and mRNA splicing (i.e., SF1, **Figure 5B**, D). Interestingly, WNT/ β -catenin transcriptional regulators were identified bound to GATA6, particularly at day 2 of cardiac differentiation, including the β -catenin transcriptional co-activator EP300⁵⁴, and the TCF/LEF transcriptional co-repressors CTBP1/2^{55,56} and TLE1⁵⁷ (**Figure 5B**), suggesting that GATA6 may directly modulate the transcriptional response induced by WNT signaling by binding to these proteins.

Proteins identified by GATA6-RIME only at day 2 included the transcription factors EOMES and MIXL1 (both transcription factors involved in primitive streak formation and mesoderm patterning^{37,58}) as well as TLE1, EP300 and SMARCA5, while the unique day 4 proteins were all RNA binding proteins such as TAF15 (**Figure 5C**). The overlapping day 2 and day 4 GATA6-RIME results included many chromatin remodelers such as SMARCC1, SMARCA4 and KDM1A (**Figure 5C**), although the enrichment for these proteins was greater at day 2 compared to day 4 (**Figure 5B**, Supplemental Table 1). As day 2 of differentiation had the highest level of GATA6 expression and enrichment for RIME identified proteins, we performed GATA6 immunoprecipitation (IP) on *GATA6*^{+/+} and *GATA6*^{-/-} samples at day 2 and confirmed a weak interaction with EOMES and SMARCC1 in the WT samples, with no observable bands in *GATA6*^{-/-} samples (**Figure 5E**). The EOMES and SMARCC1 protein bands identified in GATA6-IP samples were notably less than observed for the input controls, therefore it's possible that the interaction between GATA6 and these proteins is indirect (i.e., GATA6 may be co-bound in a complex with EOMES and/or SMARCC1 among other proteins).

Previous studies performed GATA6-RIME at day 4 and EOMES-RIME at day 2 of DE differentiation^{21,43}, so we compared our day 2 and day 4 GATA6-RIME data to these previously published RIME datasets (Supplemental Figure 6B). 26 proteins were present in all 4 datasets (Supplemental Figure 6B, C), implying there is a common interaction between GATA6 and various remodeler proteins, the transcription factor SALL4, and the TCF/LEF co-repressors CTBP1/2 during mesendoderm patterning toward both cardiac mesoderm and DE lineages. Protein interactions unique to day 2 GATA6-and EOMES-RIME included MIXL1, SMARCC2, and ARID1B and may represent protein interactions specific to mesendoderm development (Supplemental Figure 6C). Notably, EOMES and the WNT-related proteins TLE1 and EP300 were not identified in our day 4 dataset but were present in our day 2 GATA6-RIME data and the published EOMES and GATA6-RIME datasets during early DE development^{21,43}. Therefore, an interaction between GATA6 and these proteins appears to be relinquished by the cardiac mesoderm stage (day 4) but persists during DE patterning. Cardiac differentiation requires a transient inhibition of the canonical WNT

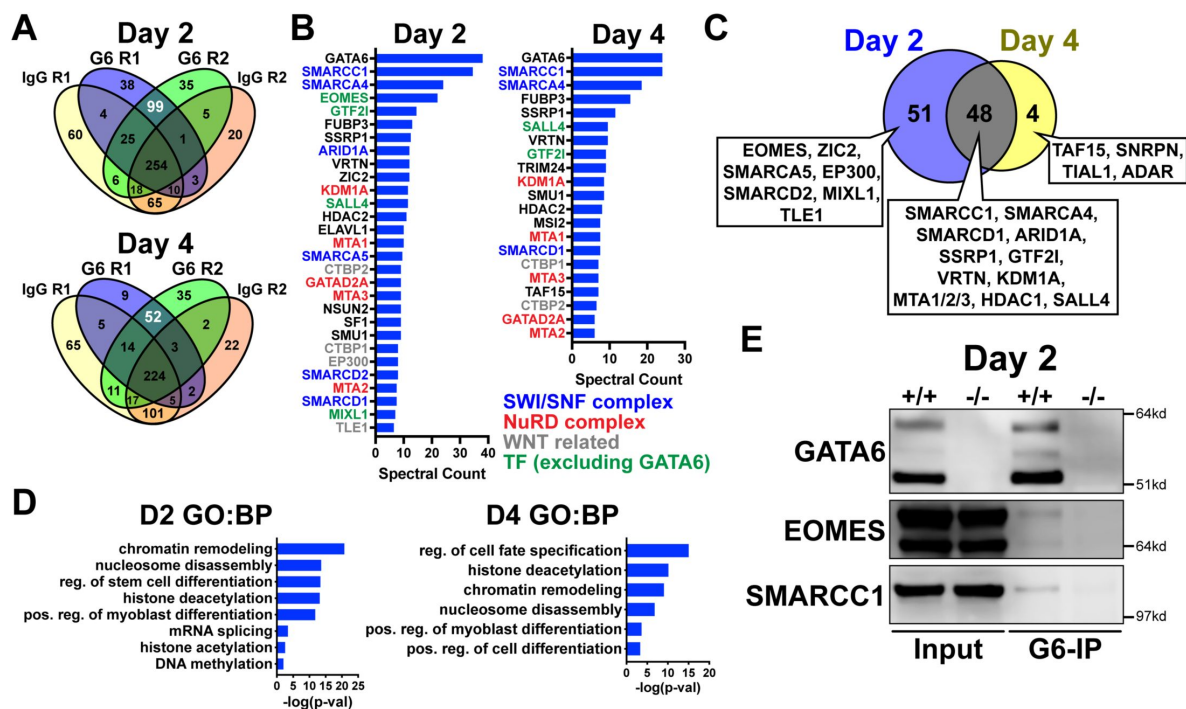


Figure 5.

GATA6 interactome analysis during precardiac to cardiac mesoderm patterning stages.

(A) Venn diagrams showing unique proteins identified by RIME analysis performed on *GATA6*^{+/+} hESCs at day 2 or day 4 of cardiac differentiation. Numbers in white text indicate enriched proteins identified by GATA6-RIME. G6 indicates GATA6, R indicates replicate (n=2). (B) Enriched proteins identified by GATA6 RIME (spectral count >5, unique peptides >1). (C) Venn diagram comparing day 2 (blue) and day 4 (yellow) GATA6-RIME enriched proteins. (D) GO (BP) analysis for GATA6-RIME enriched proteins on days 2 or 4. (E) Western blots for GATA6, EOMES, and SMARCC1 performed on GATA6-immunoprecipitated (G6-IP) whole-cell protein lysates from *GATA6*^{+/+} and *GATA6*^{-/-} cells isolated at day 2 of cardiac differentiation. Input indicates whole-cell protein lysate controls.

signaling pathway⁵⁹, and as day 4 of our cardiac differentiation protocol occurs after WNT inhibition these varying protein interactions may reflect the differing contributions of the WNT pathway during cardiac and DE differentiation.

Early manipulation of the WNT and BMP pathways partially rescues the CM defects in *GATA6* mutant hESCs

LGR5, a component of the WNT pathway⁶⁰, was among the most significant dDEGs identified from day 2 or day 3 RNA-seq in *GATA6*^{-/-} cells (relative to WT) and was also found significantly bound by *GATA6* (and in a previous study, *EOMES*⁴⁹) in WT cells (Supplemental Figure 5). As *LGR5* is required for efficient CM differentiation in hPSCs^{50,61}, we hypothesized that infection with a doxycycline (DOX) inducible *LGR5*-expressing lentivirus might rescue the cardiac defects in *GATA6*^{-/-} hESCs. In *GATA6*^{-/-} hESCs that had been transduced with the inducible-*LGR5* lentivirus (*iLGR5*), DOX was added from days 1 to 4 of cardiac differentiation using the cytokine-based protocol (Figure 6A) to mimic the *LGR5* expression pattern (Supplemental Figure 7A); we validated that this increased *LGR5* expression in a DOX-dose-dependent manner (Supplemental Figure 7B). In all differentiation experiments using *iLGR5* treated with DOX (250ng/mL), we did not observe generation of beating CMs. However, DOX treatment of *iLGR5*-transduced *GATA6*^{-/-} hESCs did yield a small but significant increase in the %K⁺P⁺ population at day 5 of differentiation relative to empty vector (EV) transduced controls, associated with increased *KDR* expression (Figure 6B). Thus, *LGR5* expression can partially rescue a proportion of day 5 cardiac mesoderm in *GATA6*^{-/-} hESCs but cannot sufficiently improve subsequent CM differentiation.

As *LGR5* is a single component of the complex WNT pathway, we tested a broader pharmacological approach by using the WNT agonist CHIR to determine if early WNT activation could more efficiently rescue the cardiac defects in *GATA6*^{-/-} hESCs. We found that inclusion of CHIR (3μM) from days 0 to 2 in differentiating *GATA6*^{-/-} hESCs yielded by day 13 small clusters of beating CMs (Supplemental Video 1). When assessing quantitatively by flow cytometry the %cTnT⁺ CMs in CHIR-treated *GATA6*^{-/-} cells relative to vehicle treated controls, there was a variable trend for CM induction that was however not statistically significant (Supplemental Figure 7C). Similarly, while the %K⁺P⁺ cardiac mesoderm cells at day 5 was slightly increased upon CHIR treatment in *GATA6*^{-/-} cells, this was not a statistically significant improvement (Supplemental Figure 7D).

BMP-related gene expression was also dysregulated at day 2 in *GATA6*^{-/-} cells and thus manipulation of the BMP4 concentration in combination with CHIR treatment from days 0 to 2 was tested to further promote CM rescue. Interestingly, we found that decreasing the BMP4 concentration (5ng/mL) combined with CHIR treatment yielded small clusters of beating CMs (Supplemental Video 2) more consistently than increasing the BMP4 concentration (data not shown) or treatment with CHIR on its own. Quantification by flow cytometry for %cTnT⁺ CMs in *GATA6*^{-/-} cells treated with CHIR and the lower BMP4 concentration (CHIR LB) showed a small but significant increase (~4% cTnT⁺) relative to *GATA6*^{-/-} controls indicating that CM differentiation is partially rescued upon CHIR LB treatment (Figure 6C). Note that this treatment actually decreases efficiency for differentiation in the control WT cells. The %K⁺P⁺ and %K⁺ day 5 cardiac mesoderm population was also significantly increased with CHIR LB treatment in *GATA6*^{-/-} cells relative to untreated controls and was not significantly different than WT controls indicating a rescue of day 5 K⁺P⁺ cardiac mesoderm (Figure 6D). Taken together, these results indicate that modulating the early input of the WNT and BMP pathways in *GATA6*^{-/-} cells can rescue the cardiac mesoderm patterning defects and partially rescue subsequent CM differentiation.

GATA6^{+/-} cells also exhibited partial dysregulated gene expression related to WNT signaling via day 2 RNA-seq relative to WT controls (Supplemental Figure 4G). We therefore hypothesized that the addition of CHIR from days 0 to 2 in differentiating *GATA6*^{+/-} hESCs could rescue CM differentiation more potently than observed in *GATA6*^{-/-} hESCs. Indeed, we found that the %cTnT⁺ CMs was significantly increased in *GATA6*^{+/-} cells supplemented with CHIR relative to vehicle

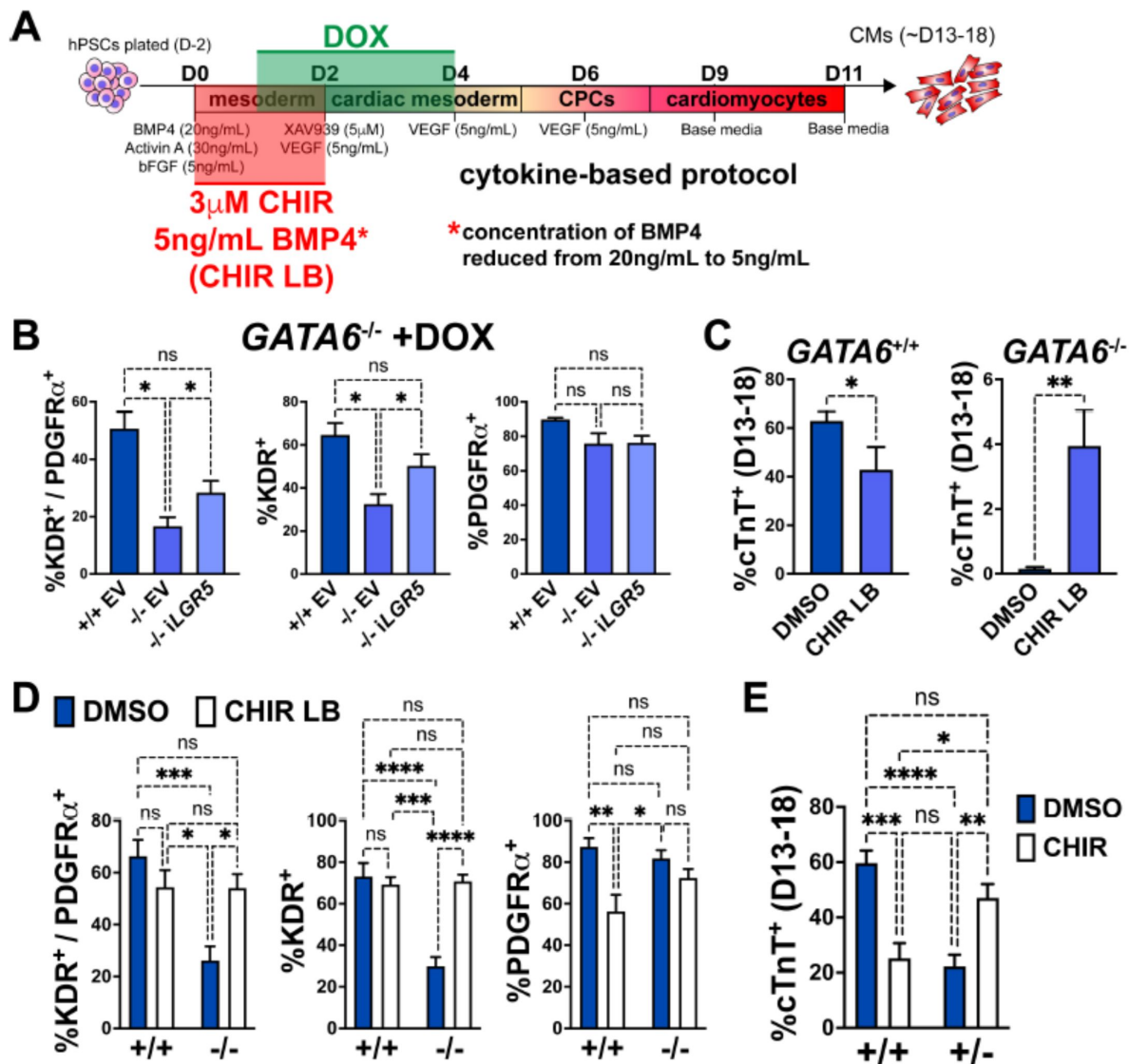


Figure 6.

Early manipulation of the WNT and BMP pathways partially rescues the CM defects in *GATA6* loss-of-function hESCs.

(A) Schematic for treatment with DOX (days 1-4), CHIR (3μM, days 0-2), and/or reduced BMP4 (5ng/mL, days 0-2, indicated as "LB") during CM directed differentiation using the cytokine-based protocol. (B) Day 5 flow cytometry quantification for %K⁺P⁺ double-positive cells, %KDR⁺ single-positive cells, or %PDGFRα⁺ single-positive cells in WT or *GATA6*^{-/-} hESCs transduced with *iLGR5* or empty vector (EV) (n=5). Significance indicated by *p<0.05 according to one-way anova and Tukey's post-hoc analysis. (C) %cTnT⁺ CMs from days 13-18 of cardiac differentiation quantified by flow cytometry in WT or *GATA6*^{-/-} cells treated with CHIR LB or vehicle (DMSO) with normal BMP4 concentration treated control (n≥6). (D) Flow cytometry at day 5 of cardiac differentiation to quantify %K⁺P⁺ double-positive, %KDR⁺ single-positive, or %PDGFRα⁺ single-positive cells comparing WT or *GATA6*^{-/-} hESCs treated with CHIR LB with *GATA6*^{+/+} and *GATA6*^{-/-} hESCs controls treated with vehicle and normal BMP4 concentration (n≥8). (E) %cTnT⁺ CMs from days 13-18 of cardiac differentiation quantified by flow cytometry in *GATA6*^{-/-} or WT hESCs treated with CHIR (3μM) or DMSO (n≥7). Data represents the mean ± SEM, significance indicated by **p<0.01, ***p<0.001, ****p<0.0001 by two-tailed Student's t-test (C) and two-way ANOVA (D, E) with Tukey's multiple comparisons test.

treated controls and was not significantly different from WT untreated controls, indicating an efficient rescue of CM differentiation (**Figure 6E**). CHIR addition to WT hESCs had a deleterious effect on CM differentiation efficiency, with the %cTnT⁺ CMs reduced from ~60% (DMSO treated) to ~25% (CHIR treated, **Figure 6E**), suggesting that there is a threshold “window” for the degree of WNT pathway activation required during early primitive streak to mesoderm patterning stages and that reducing or exceeding this threshold has an adverse effect on later CM differentiation capacity. Furthermore, this data indicates that the rescue in CM differentiation efficiency in *GATA6*^{+/-} cells treated with CHIR was not due to an overall improvement in the general cardiac differentiation protocol, but rather related to correcting the consequence of loss of one *GATA6* allele.

Discussion

We found that *GATA6* loss-of-function mutations severely disrupt CPC gene expression and impair CM differentiation, consistent with the results described by Sharma et al.¹⁹. Here, we discovered that *GATA6* is required very early in this process and that the later cardiac phenotypes observed in *GATA6*^{+/-} and *GATA6*^{-/-} hESCs originate from an early mesoderm patterning defect in which lateral and cardiac mesoderm genes fail to be expressed at normal levels. Gene networks induced by the WNT and BMP pathways that are normally required for primitive streak organization and subsequent anterior lateral mesoderm emergence are also mis-expressed when *GATA6* is deficient. *GATA6* CUT&RUN analysis revealed binding at distal enhancer-like signatures nearby dDEGs identified by RNA-seq, and *GATA6* binding was found to have a considerable overlap with previously published EOMES CHIP-seq data during early DE differentiation⁴⁹ nearby many of these genes. Day 2 *GATA6* RIME analysis confirmed binding with EOMES and various chromatin remodelers including components of the SWI/SNF and NuRD remodeling complexes that are known to regulate chromatin accessibility of cardiac lineage-specific gene networks^{62–66}. Furthermore, we found that early manipulation of the WNT pathway through CHIR treatment from days 0 to 2 rescued CM defects observed in *GATA6*^{+/-} hESCs while CHIR treatment and reduced BMP4 dosage rescued K⁺P⁺ cardiac mesoderm and partially rescued CM differentiation in *GATA6*^{-/-} hESCs. Together, these results present multiple novel functions for *GATA6* in regulating early precardiac mesoderm patterning during human cardiac differentiation.

WNT and BMP related genes were dysregulated in *GATA6*^{+/-} and *GATA6*^{-/-} cells at day 2 and day 3 of cardiac differentiation but β -catenin and SMAD protein activation was intact indicating that the transcriptional response of these pathways is defective rather than the upstream activation induced by morphogens. *GATA6* RIME analysis revealed an interaction with several WNT transcriptional regulators including the β -catenin co-activator EP300 and the TCF/LEF transcriptional co-repressors TLE1 and CTBP1/2, and binding to these proteins could have positive or negative effects on their function. One potential mechanism is that *GATA6* binds to these WNT regulators and prevents their repressive (or in the case for EP300, potentiating) function on TCF-mediated transcription, causing a net increase in the β -catenin transcriptional response. Indeed, there was an enrichment for negative regulation of the WNT signaling pathway at day 2 in *GATA6*^{-/-} cells according to RNA-seq data, suggesting that *GATA6* loss-of-function increases feedback inhibition of TCF/LEF-mediated transcription. *GATA6* CUT&RUN analysis also revealed binding near many genes related to WNT and BMP signaling implying direct participation in modulating appropriate β -catenin/SMAD transcriptional responses. However, many WNT/BMP response genes dysregulated in *GATA6*^{+/-} and *GATA6*^{-/-} cells were not identified as targets by *GATA6* CUT&RUN, suggesting that another method for regulation is indirect, perhaps related to the interaction with chromatin remodelers affecting DNA accessibility.

RIME analysis revealed *GATA6* bound to many chromatin remodelers including components of the SWI/SNF (e.g., SMARCA4) and NuRD (e.g., KDM1A) complexes at days 2 and 4 of cardiac differentiation. *GATA6* is a pioneer transcription factor with specialized function to bind and

promote accessibility of chromatin during CPC development¹⁹, and GATA6, GATA4, and GATA3 have previously been shown to interact with various members of SWI/SNF complexes such as SMARCA4^{19,21,67,68}. However, the molecular mechanism for how GATA6 modifies the accessibility of chromatin and mobilizes nucleosomes is unclear. The AP-1 pioneer factor interacts with SWI/SNF complexes to promote their localization to non-permissive regions of chromatin⁶⁹. GATA6 may function similarly, for example, by binding to SWI/SNF complex proteins and recruiting them to regions of closed chromatin containing lineage-specific enhancers where they then function to promote genomic accessibility. It is somewhat surprising that among all the SWI/SNF factors, the one that is known to be cardiac-specific (SMARCD3) was not found in the GATA6-RIME-enriched proteins. We note that SMARCD3 is expressed in the RNA-seq data as early as day 2. Although speculative, it may be that GATA6 primarily interacts with other SWI/SNF complexes prior to a subsequent role for SMARCD3 in cardiac specification.

The function of GATA6 during early mesoderm patterning presented in this study has unique implications toward understanding the pathogenesis of CHD and extra-cardiac phenotypes associated with human *GATA6* mutations. *GATA6* homozygous loss-of-function mutations have not been described in humans, most likely due to early embryonic lethality causing non-viable embryogenesis. This is consistent with the early mesoderm patterning defects of *GATA6*^{-/-} hESCs described in this study, previous reports of DE defects in *GATA6*-null hPSC lines^{20–23}, as well as the early lethality reported in *GATA6* KO mice⁷. In the case of haploinsufficiency, CHD presenting as OFT or septal defects associated with heterozygous *GATA6* mutations may be due to defects that occur during early mesoderm patterning that diminish the later developmental pool of CPCs and CMs. Quite remarkably, profiles of cells at day 2 of differentiation already identify OFT morphogenesis as gene sets disturbed by loss of GATA6. Relatively minor changes in early mesoderm cell population numbers have recently been reported to associate with later heart defects in a mouse model of Cornelia de Lange syndrome using *NPBL1*^{+/-} mice⁷⁰, and thus *GATA6* heterozygous mutations may similarly have a subtle negative effect on developing mesodermal cell populations to cause CHD. Human mutations in *GATA6* commonly associate with OFT defects, a primarily second heart field (SHF) derived structure, and recent work has established that SHF lineage progenitors are derived from early defined mesodermal subpopulations²⁸. Therefore, *GATA6* mutations might impair specific heart field progenitor populations based on the timing and positioning of these developing mesodermal populations in the primitive streak, long before the progenitors are associated with SHF. *GATA6* haploinsufficiency in humans also associates with endoderm defects such as pancreatic agenesis, and DE and early mesoderm co-develop from a common mesendoderm population adjacent to one another in the anterior primitive streak during gastrulation⁷¹. Defects in the regulatory functions of GATA6 during mesendoderm patterning, such as interactions with EOMES and specific chromatin remodelers, may cause downstream organ defects in both mesoderm and endoderm derived organ systems in some patients with *GATA6* mutations. The data presented here focuses on early stages of mesoderm patterning and cardiac progenitor differentiation in the context of *GATA6* loss-of-function rather than CM profiling and is thus a limitation of our study. Future investigation of how heterozygous *GATA6* mutations impact CMs physiology such as cardiac chamber identity and CM function would be a worthwhile pursuit to expand our knowledge of how *GATA6* regulates CM differentiation in relation to human CHD.

Materials and Methods

hPSC Culture

hPSC lines were cultured on Matrigel (Corning BD354277) coated tissue culture plates using StemFlex (Thermo Fisher Scientific A3349401) or mTESR Plus (Stem Cell Technologies 1000276) medium at 37°C and 5% CO₂. Cells were passaged using Accutase Cell Detachment Solution (VWR

10761-312) and supplemented with 10 μ M ROCK inhibitor (RI) Y-27632 (Selleckchem S1049) for one day. hPSCs were routinely tested for mycoplasma contamination using the LookOut Mycoplasma PCR Detection Kit (Sigma-Aldrich MP0035).

iPSC Generation

A skin biopsy was obtained from a patient with an atrial septal defect and congenital diaphragmatic hernia associated with a heterozygous frameshift mutation in *GATA6* (c.1071delG)¹⁵. Human dermal fibroblasts (HDF) derived from this biopsy were transduced with 4 Sendai viruses expressing OCT3/4, SOX2, KLF4, and c-MYC using the CytoTune-iPS 2.0 Sendai Reprogramming Kit (Invitrogen A16517). Transduced HDFs were reprogrammed on CF1 Mouse Embryonic Fibroblasts (MEFs) (Thermo Fisher Scientific A34180) with iPSC medium consisting of DMEM/F12 (VWR 16777-255) supplemented with 20% KnockOut serum replacement (KOSR, Thermo Fisher Scientific 10828010), 1% non-essential amino acids (Thermo Fisher Scientific), 1% L-glutamine (Corning 25-005-Cl), 1% Penicillin:Streptomycin Solution (Corning 30-002-Cl), 0.0007% 2-mercaptoethanol (Thermo Fisher Scientific 21985023), 10 ng/mL bFGF (R&D Systems 233-FB-025). Once iPSC colonies emerged, colonies were manually isolated and expanded into feeder-free conditions by culturing with mTESR1 (Stem Cell Technologies 85850) on Matrigel coated plates.

CRISPR Gene Editing

The *GATA6* mutant H1 (WA01, NIH #0043) hESC cell lines were established in a previous study²⁰ using the H1-iCas9 cell line and CRISPR gene editing²⁹ targeting the C-terminal ZnF domain (Supplemental Figure 1A, B). The *GATA6* c.1071delG iPSC mutant allele was corrected to WT sequence to create the isogenic *GATA6*^{corr/+} line (as well as unmodified *GATA6*^{1071delG+} CRISPR-control clones) using CRISPR technology with Nickase-Cas9 according to previously established methods⁷². A pair of gRNA sequences were designed targeting opposing DNA strands surrounding the G deletion site. The gRNA 1 sequence was specifically designed to overlap the G deletion site to prevent Nickase-Cas9 activity on the WT *GATA6* allele. A 91 nucleotide (nt) length single-stranded DNA (ssDNA) repair template (Integrated DNA Technologies) was designed with 45 nt homology arms flanking the G mutation site, including a G to A silent mutation in the right homology arm and gRNA 2 recognition sequence to prevent Nickase-Cas9 activity after homology directed repair (HDR). gRNAs 1 and 2 (Supplemental Table 2) were cloned into the pSpCas9n(BB)-2A-Puro (PX462) V2.0 plasmid (Addgene 62987) as described previously⁷². 2 million dissociated *GATA6* c.1071delG/+ iPSCs were electroporated with 4 μ g each gRNA-PX462 vector with 3 μ L of 100 μ M/L ssDNA donor oligonucleotide using the Neon Transfection System (2 pulses at 1450V for 20 milliseconds, Invitrogen NEON1S) and plated on Matrigel coated dishes with mTESR Plus supplemented with 10 μ M RI. A T7 endonuclease I assay (New England Biolabs M0302S) was performed to quantify indel frequency of gRNA pairs following PCR amplification (Phusion Hot Start Flex DNA Polymerase, New England Biolabs M0535L) and agarose gel electrophoresis from genomic DNA isolated 2-3 days after electroporation using the DNeasy Blood & Tissue Kit (Qiagen 69506). 24 hr after electroporation, the culture medium was replaced with mTESR Plus supplemented with 0.5 μ g/mL Puromycin (Sigma-Aldrich P8833) for antibiotic resistance selection for 48 hr. iPSCs were then dissociated with Accutase and plated at a low density (~3000 iPSCs per 10cm plate) on Matrigel coated dishes with mTESR Plus medium (with 10 μ M RI for the first 24 hr) and allowed to grow for about 2 weeks before iPSC colonies were manually isolated and plated into Matrigel coated 96 well plates and cultured with mTESR Plus. DNA was isolated from iPSC clones using QuickExtract DNA Extraction Solution (Biosearch Technologies QE09050), PCR amplification was performed using primers surrounding the mutation site (Supplemental Table 2) and Sanger sequencing was performed to verify gene editing (GENEWIZ). Karyotyping was performed by WiCell Research Institute (Madison, WI).

hPSC Cardiomyocyte Differentiation

A monolayer cytokine-based cardiac differentiation protocol was adapted from previously established methods (Figure 1A^{26,27}). Briefly, a “base medium” for cardiac differentiation was prepared using RPMI 1640 (Thermo Fisher Scientific 22400089) supplemented with 0.5x B27 (Thermo Fisher Scientific 17504044), 0.5 mM 1-Thioglycerol (Sigma-Aldrich M6145), 50 µg/mL L-Ascorbic acid (Sigma-Aldrich A4544), and 1x GlutaMAX (Thermo Fisher Scientific 35050061). The hPSCs were plated at $\sim 4 \times 10^4$ cells/cm² and allowed to grow for 2 days before changing with “Day 0 medium” consisting of the “base medium” plus 20 ng/mL BMP4 (R&D Systems 314-BP-050), 30 ng/mL Activin A (R&D Systems 338-AC-050), 5 ng/mL bFGF (R&D Systems 233-FB-025), and 5 µL/mL Transferrin (Roche 10652202001). BMP4 was used at 10 ng/ml for iPSC differentiation. Approximately 55 hr later, the medium was changed with “Day 2 medium” consisting of “base medium” supplemented with 5 µM XAV-939 (Sigma-Aldrich X3004), 5 ng/mL VEGF (R&D Systems 293-VE-050), and 5 µL/mL Transferrin. The culture medium was changed on days 4 and 6 of cardiac differentiation with “base medium” supplemented with 5 ng/mL VEGF. From day 9 and later the culture medium was changed with “base medium” every 2-3 days. For a subset of experiments, 3 µM CHIR99021 (Stem Cell Technologies 72054) was included in the “Day 0 medium” and/or the concentration of BMP4 was reduced to 5 ng/mL (Figure 6C-E, Supplemental Figure 7C, D). An alternative CM differentiation protocol based on chemical manipulation of the WNT pathway³⁰ was used for a subset of experiments (CHIR protocol, Supplemental Figure B-E). For the CHIR protocol, hESCs were plated at $\sim 4 \times 10^4$ cells/cm² and allowed to grow for 2 days in mTESR Plus before changing medium to RPMI 1640 with 1x B27 minus insulin (Thermo Fisher Scientific A1895601) and 6 µM CHIR99021 (Stem Cell Technologies 72054) for 24 hr. Medium was then changed to RPMI 1640 with 1x B27 minus insulin for another 24 hr, before changing medium on day 3 to RPMI 1640 with 1x B27 minus insulin and 5 mM IWP2 (Tocris 3533). On day 5, culture medium was changed to RPMI 1640 with 1x B27 minus insulin once more and starting on day 7 medium was changed to RPMI with 1x B27 every other day. CMs were purified using a lactate-selection protocol to prepare for immunocytochemistry by changing the differentiation medium on day 15 of the cytokine-based protocol to a lactate-selection medium consisting of RPMI 1640 lacking glucose (Thermo Fisher Scientific 11879020) supplemented with 5 mM Sodium L-lactate (Sigma-Aldrich L7022), 213 µg/mL L-ascorbic acid, and 500 µg/mL Human Albumin (Sigma-Aldrich A9731). Lactate-selection medium was refreshed on day 17 of differentiation, after which the medium was restored to “base medium” on day 19 and changed every 2 to 3 days prior to fixation.

Flow Cytometry

Cells were detached from culture plates using 0.5% Trypsin / 0.5mM EDTA (VWR 45000-664) or TrypLE Express Enzyme (Thermo Fisher Scientific 12605010) and resuspended in a FACS buffer consisting of PBS with 10% FBS (for cTnT staining) or 0.5% BSA (for live cell staining). Cells were fixed by pelleting at 300g and resuspended in 0.2% paraformaldehyde (PFA) for 25 min at RT before staining with primary antibody for cTnT (Thermo Fisher Scientific MA5-12960) in FACS buffer with 0.5% Saponin (Sigma-Aldrich S7900-25G) for 1 hr at RT, washed 3x in FACS buffer, and then stained for 1 hr at RT with FACS buffer plus Saponin containing secondary antibody (Invitrogen A21236). Live cells were used for staining the cell surface proteins KP by incubating with FACS buffer containing PE-conjugated KDR antibody (R&D Systems FAB357P-100) and APC-conjugated PDGFRα antibody (R&D Systems FAB1264A) or IgG control antibodies (R&D Systems IC002P, IC002A) for 30 min at RT, and then stained with SYTOX Green Dead Cell Stain (Thermo Fisher Scientific S34860) in FACS buffer for 5 min at RT. BRACHYURY flow cytometry was performed using the Foxp3 Transcription Factor Staining Buffer Set (eBioscience 00-5523-00) according to the manufacturer's protocol using a PE-conjugated BRACHYURY antibody (R&D Systems IC2085P) or IgG control antibody (R&D Systems IC108P). Flow cytometry was performed and analyzed using an Attune NxT Flow Cytometer (ThermoFisher Scientific) for fixed cells and a BD-Accuri C6 Flow Cytometer (BD Biosciences) for live cells.

RT-qPCR

RNA was isolated using the RNeasy Mini Kit (QIAGEN 74106) and reverse-transcribed using the SuperScript VILO cDNA Synthesis Kit (Invitrogen 11754250) according to the manufacturer's protocols. The qPCR reaction was prepared by mixing cDNA with LightCycler 480 SYBR Green Master mix (Roche 04887-352-001) and primers listed in Supplemental Table 2. RT-qPCR was performed using a LightCycler 480 Instrument II (Roche). Gene expression data was normalized to endogenous *HPRT* expression levels as a control for each sample.

Western Blotting

Whole cell lysates were prepared using RIPA Lysis Buffer (Thermo Fisher Scientific 89900) supplemented with 1x Protease/Phosphatase Inhibitor Cocktail (Cell Signaling Technology 5872S) and 1x Benzonase (Millipore 70664-3). Protein was quantified using the Pierce BCA Protein Assay Kit (Thermo Fisher Scientific 23225) and electrophoresed on NuPAGE 4 to 12% Bis-Tris Gels (Invitrogen NP0335BOX) before transferring to PVDF membranes (Bio-Rad 162-0177). Blots were washed with 1x TBS with 0.1% Tween-20 (TBST) and blocked using 5% IgG-free BSA (VWR RLBSA50). Primary antibodies (Supplemental Table 2) were added to blocking buffer and stained overnight at 4°C. Secondary antibody staining was performed for 1 hr at RT using HRP-conjugated antibodies (Supplemental Table 2) in blocking buffer. Pierce™ ECL Western Blotting Substrate (Thermo Fisher Scientific 32106) used used to visualize proteins, and imaging was performed using a C-DiGit Blot Scanner (LICOR) with ImageStudio (v5.2) and Adobe Photoshop (v25.9.1) software.

Co-Immunoprecipitation

Co-immunoprecipitation (Co-IP) was performed using the Pierce™ Crosslink Magnetic IP/Co-IP Kit (Thermo Fisher Scientific 88805) according to the manufacturer's protocol using cells at day 2 of cardiac differentiation. 7μg of the GATA6 antibody (Cell Signaling Technology 5851) was used to cross-link to the Protein A/G Magnetic beads and 1 mg of protein was used for the IP reaction performed overnight at 4°C on a rotating platform. GATA6 Co-IP samples were electrophoresed on a gel and imaged as described for Western Blotting using 30 μg of protein lysate from the Co-IP used as an input control. *GATA6*^{-/-} cell samples served as a negative control.

Immunocytochemistry and Microscopy

Cells were fixed in tissue culture dishes with 4% PFA for 20 min at RT. Cells were washed in PBS and then blocked for 1 hr at RT with blocking buffer consisting of PBS with 10% FBS, 0.1% IgG-free BSA (Rockland Immunochemicals BSA-10), 0.1% Saponin, and 10% goat serum (Jackson ImmunoResearch Laboratories Inc. 005-000-121). Cells were then stained with primary antibodies (Supplemental Table 2) diluted in blocking buffer (without goat serum) overnight at 4°C. Secondary antibody staining was performed in blocking buffer (without goat serum) for 1 hr at RT. Cells were stained with DAPI to visualize nuclei and imaged using a Zeiss LSM800 laser scanning confocal microscope with ZEN imaging software or a Zeiss Epifluorescence microscope with AxioVision software, and edited using Adobe Photoshop software (v25.9.1). Videos were taken using an Infinity5 Microscope Camera (Teledyne Lumenera) with Infinity Analyze software.

Lentivirus Production and Infection

LGR5 full length cDNA was first cloned into a DOX-inducible Lentiviral vector containing a GFP selection marker (CS-TRE-PRE-Ubc-tTA-I2G⁷³) by performing PCR for *LGR5* using WT cDNA samples from day 2 of cardiac differentiation and primers designed to incorporate the BsiWI and MfeI restriction enzyme recognition sites flanking *LGR5* (Supplemental Table 2) to allow digestion and insertion into the CS-TRE-PRE-Ubc-tTA-I2G vector. *LGR5* sequence was validated by Sanger sequencing (GENEWIZ) using sequencing primers listed in Supplemental Table 2. Lentivirus

production was performed by transfecting 293T cells with the lentiviral expression vectors (LGR5 or EV control) and packaging plasmids using PEI. Transfected 293T cells were then allowed to produce lentivirus for 48 hr before the medium was collected, and the virus was concentrated using Lenti-X Concentrator (Takara 631231) according to the manufacturer's protocol and resuspended in StemFlex medium. *GATA6*^{-/-} hESCs were infected with lentivirus by replacing medium with *iLGR5* or EV control⁷⁴ lentivirus concentrate in StemFlex for 16 hr before changing to normal StemFlex and seeded for 2 to 3 days. Lentivirus infected hESCs were then sorted based on GFP expression using FACS through the Weill Cornell Medicine Flow Cytometry Core Facility. Doxycycline (DOX, Sigma-Aldrich D9891) was added from days 1 to 4 to induce expression of *iLGR5* or EV infected *GATA6*^{-/-} hESCs.

RNA-seq

Total RNA was extracted from hESCs approximately 52 hr (day 2), 74 hr (day 3), or 5 days after starting the cardiac differentiation protocol using the RNeasy Mini Kit (QIAGEN 74106) from at least two independent biological replicates per sample. Total RNA integrity was verified using a 2100 Bioanalyzer (Agilent Technologies), and RNA concentrations were measured using the NanoDrop system (Thermo Fisher Scientific). Libraries were generated from day 2 and day 5 RNA using the TruSeq RNA Library Prep Kit v2 (Illumina RS-122-2001) and day 3 RNA using the TruSeq Stranded mRNA Library Prep Kit (Illumina 20020594) according to the manufacturer's instructions. Normalized cDNA libraries were pooled and sequenced on Illumina HiSeq4000 (Day 2 and Day 5, single read) or Illumina NovaSeq 6000 (Day 3, paired-end read) sequencing systems for 50 cycles through the Weill Cornell Medicine Genomics Resources Core Facility. The raw sequencing reads in BCL format were processed through bcl2fastq 2.19 (Illumina) for FASTQ conversion and demultiplexing. Day 2 and Day 5 RNA-seq alignment (GRCh19) and differential gene expression analysis was performed as described⁷⁵. Day 3 RNA-seq reads were aligned and mapped to the GRCh38 human reference genome by STAR (Version 2.5.2⁷⁶) and transcriptome reconstruction was performed by Cufflinks (Version 2.1.1, <http://cole-trapnell-lab.github.io/cufflinks/>) after trimming adaptors with cutadapt (Version 1.18, <https://cutadapt.readthedocs.io/en/v1.18/>). The abundance of transcripts was measured with Cufflinks in Fragments Per Kilobase of exon model per Million mapped reads (FPKM)^{77,78}. Raw read counts per gene were extracted using HTSeq-count v0.11.2⁷⁹. Gene expression profiles were constructed for differential expression, cluster, and principle component analyses with the DESeq2 package⁸⁰. For differential expression analysis, pairwise comparisons between two or more groups using parametric tests where read-counts follow a negative binomial distribution with a gene-specific dispersion parameter were performed. Corrected p-values were calculated based on the Benjamini-Hochberg method to adjust for multiple testing. GSEA was performed using GSEA 4.2.3 software (Broad Institute, Inc. and Regents of the University of California)⁸¹ on RNA-seq gene lists ranked by log₂(Fold Change) using the subcollection of biological process (BP) gene sets from the GO collection in the Human Molecular Signatures Database (MSigDB)⁸², or using custom gene lists from published sequencing datasets as described in this manuscript^{22,40,44}, and gene sets with a FDR (False Discovery Rate) *q*-value < 0.25 and a nominal *p*-value < 0.05 were considered to be enriched significantly. GO analysis was performed using the Database for Annotation, Visualization, and Integrated Discovery (DAVID, david.ncifcrf.gov)^{83,84} using the BP subcollection. Heatmaps were generated from normalized counts from selected gene lists using SRplot (bioinformatics.com.cn/srplot)⁸⁵. Differentially expressed genes (DEGs) were defined as *p*-value adjusted (*p*-adj) < 0.05 and log₂(fold change) > 0.5, genes with a count below 50 for all samples were excluded.

CUT&RUN

500,000 H1-*GATA6*^{+/+} cells were harvested approximately 52 hrs after starting the cardiac differentiation protocol and CUT&RUN was performed using the CUTANA ChIC/CUT&RUN Kit (Epicpyher 14-1048) according to the manufacturer's protocol. Anti-*GATA6* (n=3 independent biological replicates) (Cell Signaling Technology 5851) and Rabbit Anti-IgG (n=2 independent

biological replicates) negative control (Epicypther 13-0042) antibodies were used at a 1:50 dilution (~0.5µg). CUT&RUN library preparation, sequencing, and SEACR-based analysis were performed by the Weill Cornell Medicine Epigenomics and Genomics Core Facilities. The DNA libraries were prepared with approximately 5 ng of CUT&RUN enriched DNA using the NEBNext Ultra II DNA Library Prep Kit for Illumina (New England Biolabs E7645) and NEBNext Multiplex Oligos for Illumina (96 Unique Dual Index Primer Pairs) (New England Biolabs E6440); 14 PCR cycles were used to amplify the adaptor ligated target DNA. The libraries were clustered on a pair end flow cell and sequenced for 50 cycles on an Illumina NextSeq500 Sequencing System. FASTQ files were aligned to the GENCODE GRCh38 build of the human genome using bowtie2 with the `—dovetail` parameter. The resulting BAM files were sorted and indexed using samtools and then converted to bedgraph format using bedtools. Peaks were subsequently called using SEACR to identify enrichment in target data by selecting the top 1% of regions by AUC (stringent threshold). The DiffBind R Bioconductor package was used to interrogate differential binding for GATA6 vs IgG controls, and the annotated significant nearby TSS gene list from the SEACR analysis was used for genomic peak distribution, GO analysis (as described for RNA-seq) and venn diagram comparisons. Transcription factor motif enrichment analysis was performed using Basepair software with a CUT&RUN Peaks, Motif (MACS2, HOMER) pipeline (basepairtech.com). EOMES CHIP-seq data from Teo et al.⁴⁹ was aligned to GRCh38 using bowtie2 with Partek software (partek.com). CUT&RUN and CHIP-seq data were visualized using Integrative Genomics Viewer (IGV) 2.16.2, distal enhancer-like signatures (dELS) defined by the ENCODE project⁴⁸ were identified using the UCSC Genome Browser for hg38 (genome.ucsc.edu).

RIME Analysis

About ~50 million H1-GATA6^{+/+} cells at day 2 or day 4 of cardiac differentiation were fixed with 1% formaldehyde (Electron Microscopy Sciences 15710) for 8 min at RT with gentle agitation. Fixation was quenched with 125 mM glycine (Bio-Rad 1610718) for 5 min at RT before cells were manually scraped, transferred to conical tubes, and pelleted at 800g at 4°C for 10 min. Pellets were resuspended in PBS with 0.5% IGEPAL CA-630 (Sigma-Aldrich I8896) and centrifuged again before resuspending in PBS-IGEPAL with 1 mM PMSF (Thermo Fisher Scientific 36978) and centrifuged one final time. Supernatant was then removed from pellets which were then snap frozen on dry ice and transferred to -80°C. RIME analysis was performed by Active Motif (Carlsbad, CA) according to published methods⁸⁶ using 150 µg of chromatin per sample immunoprecipitating with anti-GATA6 (Cell Signaling Technology 5851) or anti-IgG control (Cell Signaling Technology 2729) antibodies (15 µg antibody per IP) and performing mass spectrometry with enriched GATA6 bound proteins defined as not present in the IgG controls and having a spectral count (the total number of all peptides detected) of at least 5 and >1 unique peptides.

Statistical Analysis

Data is represented as the mean ± SEM from independent biological replicates (as indicated in figure legends). Graphing and statistical analysis was performed using GraphPad Prism v.10.2.3 software with two-tailed Student's t test when comparing two groups. One-way or two-way ANOVA was performed when comparing three or more groups, unless otherwise indicated.

Data availability

All RNA-seq and CUT&RUN data is available at the GEO repository and are publicly available as of the date of publication. The series accession numbers are GSE275685 and GSE275686. Requests for primary data, resources and reagents should be directed to and will be fulfilled by the lead contact, Todd Evans (tre2003@med.cornell.edu).

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Additional files

Supplemental Figs and Tables 

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Reviewer #1 (Public review):

Summary:

This is a comprehensive study that clearly and deeply investigates the function of GATA6 in human early cardiac development.

Strengths:

This study combines hESC engineering, differentiation, detailed gene expression, genome occupancy, and pathway modulation to elucidate the role of GATA6 in early cardiac differentiation. The work is carefully executed and the results support the conclusions. The use of publicly available data is well integrated throughout the manuscript. The RIME experiments are excellent.

Weaknesses:

Much has been known about GATA6 in mesendoderm development, and this is acknowledged by the authors.

Comments on revised version:

The authors have addressed my comments appropriately.

<https://doi.org/10.7554/eLife.100797.2.sa3>

Reviewer #2 (Public review):

Summary:

This manuscript by Bisson et al describes the role GATA6 to regulate cardiac progenitor cell (CPC) specification and cardiomyocyte (CM) generation using human embryonic stem cells (hESCs). The authors found that GATA6 loss-of-function hESC exhibit early defects in mesendoderm and lateral mesoderm patterning stages. Using RNA-seq and CUT&RUN assays the genes of the Wnt and BMP programs were found to be affected by the loss of GATA6 expression. Modulating Wnt and BMP during early cardiac differentiation can partially rescue CPC and CM defects in GATA6 hetero- and homozygous mutant hESCs.

Strengths:

The studies performed were rigorous and the rationale for the experimental designed were logical. The results obtained were clear and supports the conclusions that the authors made regarding the role of GATA6 on Wnt and BMP pathway gene expression.

Weaknesses:

Given the wealth of studies that have been performed in this research area previously, the amount of new information provided in this study is relatively modest. Nevertheless, the results are quite clear and should make a strong contribution to the field.

Comments on revised version:

The authors have addressed the prior request to assess genes expression representing each stage of development/differentiation from mesoderm to cardiac progenitor to

cardiomyocytes and confirmed that the differentiation defect lies at the cardiac progenitor and cardiomyocyte stages and not in mesodermal differentiation. This work has significantly improved the robustness of the study.

<https://doi.org/10.7554/eLife.100797.2.sa2>

Reviewer #3 (Public review):

In this study, Bison et al. analyzed the role of the GATA6 transcription factor in patterning the early mesoderm and generating cardiomyocytes, using human embryonic stem cell differentiation assays and patient-derived hiPSCs with heart defects associated with mutations in the GATA6 gene. They identified a novel role for GATA6 in regulating genes involved in the WNT and BMP pathways. Modulation of the WNT and BMP pathways partially rescue early cardiac mesoderm defects in GATA6 mutant hESCs. These results provide significant insights into how GATA6 loss-of-function and heterozygous mutations contribute to heart defects.

Comments on revised version:

The authors have addressed all the concerns, using new data and modifications to the text to further strengthen the manuscript.

<https://doi.org/10.7554/eLife.100797.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This is a comprehensive study that clearly and deeply investigates the function of GATA6 in human early cardiac development.

Strengths:

This study combines hESC engineering, differentiation, detailed gene expression, genome occupancy, and pathway modulation to elucidate the role of GATA6 in early cardiac differentiation. The work is carefully executed and the results support the conclusions. The use of publicly available data is well integrated throughout the manuscript. The RIME experiments are excellent.

Weaknesses:

Much has been known about GATA6 in mesendoderm development, and this is acknowledged by the authors.

We appreciate the comments and have tried to highlight both the early role of GATA6 in cardiac progenitor biology as well as the haploinsufficiency for relevance to human congenital heart disease, which we believe adds value to other recent published work, among others Sharma et al. eLife 2020.

Reviewer #2 (Public review):

Summary:

This manuscript by Bisson et al describes the role of GATA6 to regulate cardiac progenitor cell (CPC) specification and cardiomyocyte (CM) generation using human embryonic stem cells (hESCs). The authors found that GATA6 loss-of-function hESC exhibits early defects in mesendoderm and lateral mesoderm patterning stages. Using RNA-seq and CUT&RUN assays the genes of the Wnt and BMP programs were found to be affected by the loss of GATA6 expression. Modulating Wnt and BMP during early cardiac differentiation can partially rescue CPC and CM defects in GATA6 hetero- and homozygous mutant hESCs.

Strengths:

The studies performed were rigorous and the rationale for the experimental design was logical. The results obtained were clear and supported the conclusions that the authors made regarding the role of GATA6 on Wnt and BMP pathway gene expression.

Weaknesses:

Given the wealth of studies that have been performed in this research area previously, the amount of new information provided in this study is relatively modest. Nevertheless, the results are quite clear and should make a strong contribution to the field.

Likewise for reviewer 2, we appreciate the comments and have tried to highlight both the early role of GATA6 in cardiac progenitor biology as well as the haploinsufficiency for relevance to human congenital heart disease.

Reviewer #3 (Public review):

In this study, Bisson et al. analyzed the role of the GATA6 transcription factor in patterning the early mesoderm and generating cardiomyocytes, using human embryonic stem cell differentiation assays and patient-derived hiPSCs with heart defects associated with mutations in the GATA6 gene. They identified a novel role for GATA6 in regulating genes involved in the WNT and BMP pathways -findings not previously noted in earlier analyses of GATA6 mutant hiPSCs during early cardiac mesoderm specification (Sharma et al., 2020). Modulation of the WNT and BMP pathways may partially rescue early cardiac mesoderm defects in GATA6 mutant hESCs. These results provide significant insights into how GATA6 loss-of-function and heterozygous mutations contribute to heart defects.

I have the following comments:

(1) Throughout the manuscript, Bisson et al. alternate between different protocols to generate cardiomyocytes, which creates some confusion (e.g., Figure 1 vs. Supplemental Figure 2A). The authors should provide a clear justification for using alternative protocols.

We agree and clarified this issue in the revision (p. 6). The reviewer is correct that there are two widely used protocols for directed differentiation of PSCs to cardiac fate. One is a cytokine-based protocol (Fig. 1A) and the other uses small molecules to manipulate the WNT pathway (CHIR protocol, Supplemental Fig. 2B). In our study, we used the CHIR protocol only for experiments in Supplemental Figure 2B-E. Since our data implicated BMP and WNT as mediators of the GATA6-dependent program, we did this mainly to confirm that the phenotype we observed with the cytokine-based protocol was not biased by the differentiation protocol. However, we found the CHIR protocol to be overall relatively

inefficient for cardiac differentiation using the parental H1 hESCs and the various isogenic lines. The *in vitro* cardiac differentiation protocols for hPSCs are known to be variable depending on lines and sometimes require extensive optimization for various media components and concentrations, cell seeding densities, and batch variations for crucial reagents. The cytokine-based protocol we optimized worked most efficiently with our hPSC lines to generate cardiomyocytes, therefore we committed to using it for the bulk of experiments in this study.

(2) The authors should characterise the mesodermal identity and cardiomyocyte subtypes generated with the activin/BMP-induction protocol thoroughly and clarify whether defects in the expression of BMP and WNT-related gene affect the formation of specific cardiomyocyte subtypes in a chamber-specific manner. This analysis is important, as Sharma et al. suggested a role for GATA6 in orchestrating outflow tract formation, and Bison et al. similarly identified decreased expression of NRP1, a gene involved in outflow tract septation, in their GATA6 mutant cells.

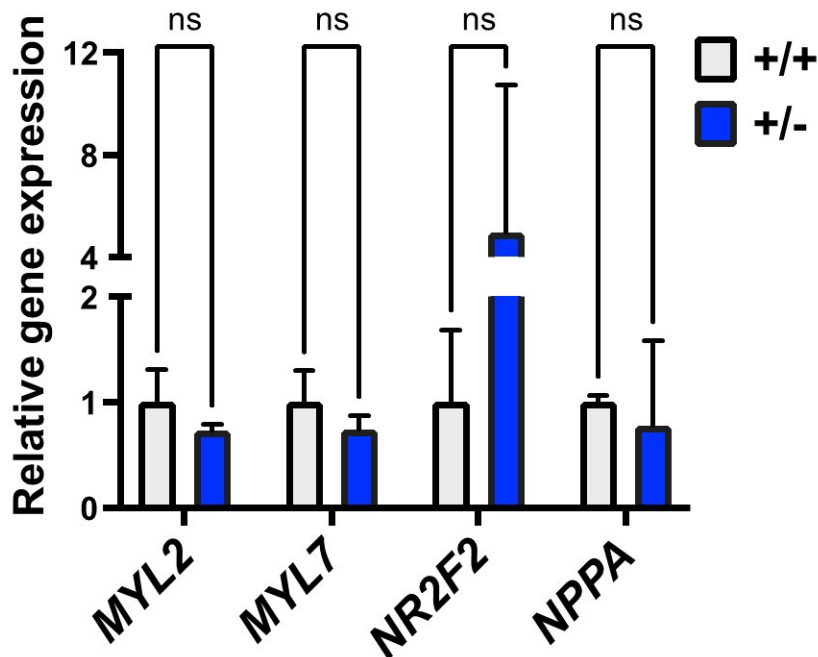
We agree it is important that the mesodermal identities are quite thoroughly characterized.

For example, Fig. 2 (K+P+, Brachyury, EOMES), Fig. 3G&H (lateral mesoderm, cardiac mesoderm RNAseq & GSEA comparing datasets from Koh et al.). The capacity of the cytokine-based protocol to generate both FHF and SHF derived sub-types has been rigorously evaluated by Keller and colleagues, which we now cite (Yang et al. 2022). Since the null cells do not generate CMs, chamber specific subtypes cannot be evaluated; whether the GATA6 heterozygous mutants are biased is an interesting question. Indeed, the top GO term identified by CUT&RUN analysis for GATA6 at day 2 of

differentiation is outflow tract morphogenesis, which is consistent with the interpretation by Sharma et al., but implicates this program at a much earlier developmental stage, long before cardiomyocyte differentiation. We think this is one of the most important findings of our study and appreciate the chance to highlight this in the revision (p. 9, 17). When we evaluated chamber-specificity for differentiated cardiomyocytes, we did not find significant differences, as indicated for the reviewer in the panel below (day 20 of differentiation). Since our study focuses on early stages of progenitor specification rather than cardiomyocyte differentiation, we agree that a more rigorous analysis would be of value, and indicated this as a limitation of our current study (p. 18).

Author response image 1.

D20 lactate-purified CMs: RT-qPCR (n=2)



(3) The authors developed an iPSC line derived from a congenital heart disease (CHD) patient with an atrial septal defect and observed that these cells generate *cTnnT*⁺ cells less efficiently. However, it remains unclear whether atrial cardiomyocytes (or those localised specifically at the septum) are being generated using the activin/BMP-induction protocol and the patient-derived iPSC line.

As indicated above, our study is focused on cardiac progenitor specification, and we found similar differences with the patient-derived iPSC-CMs compared to using hESC heterozygous targeted mutants. While we did not note any major differences in expression of cardiomyocyte markers, whether the mutants show any biases toward sub-types of cardiomyocytes is an interesting question to be pursued in subsequent work.

(4) The authors should also justify the necessity of using the patient-derived line to further analyse *GATA6* function.

This is a good point, and as suggested we provided the justification (p. 5-6). This is the first patient-derived iPSC line published with a heterozygous *GATA6* mutation along with an isogenic mutation-corrected control generated for cardiac directed differentiation. Patients with congenital heart disease (CHD) associated with *GATA6* mutations are typically heterozygous (also true for many other CHD variants; presumably homozygous null embryos would not survive). It is important to query if phenotypes found using targeted mutations in hESCs (or iPSCs) model the human disease, since the patient cells (or the hESCs) likely have additional genetic variants that might interact with the *GATA6* mutation. The fact that both

types of heterozygous cells (patient-derived iPSCs and targeted hESCs) generate similar defects in CM differentiation provides evidence supporting the use of these human cellular models to study the genetic and cellular basis for congenital heart disease. This is particularly important, since other models, such as heterozygous mice, do not show such phenotypes.

(5) Figure 3 suggests an enrichment of paraxial mesoderm genes in the context of GATA6 loss-of-function, which is intriguing given the well-established role of GATA6 in specifying cardiac versus pharyngeal mesoderm lineages in model organisms. Could the authors expand their analysis beyond GO term enrichment to explore which alternative fates GATA6 mutant cells may acquire? Additionally, how does the potential enrichment of paraxial mesoderm, rather than pharyngeal mesoderm, relate to the initial mesodermal induction from their differentiation protocol? Could the authors also rule out the possibility of increased neuronal cell fates?

We need to interpret our *in vitro* differentiation data cautiously in relation to what has been shown *in vivo*, since we are unlikely to be reproducing all the complex signaling taking place in the embryo. Yet we do see modest increases in gene expression levels including signatures of paraxial mesoderm and ECM/mesenchymal at days 2 or 3 of differentiation in the GATA6 mutant cells. Therefore, we now include a heatmap showing enriched paraxial mesoderm gene expression in the mutant cells, new Fig. 3I (see page 10).

A caveat of this result is that the cells are being differentiated toward cardiac fate, so a bias for alternative fates might be suppressed. We modified the protocol to favor paraxial fate by adding CHIR at day 2 (rather than XAV) and performing qPCR assays at day 3. We found this successfully induced paraxial mesoderm gene expression, but equally comparing wildtype, heterozygous, or null cells, so do not feel it warrants highlighting further.

Recommendations for the authors:

Reviewing Editor (Recommendations for the authors):

Incorporation of marker analysis for various stages of iPSC to CM differentiation (mesoderm, cardiac progenitor, CM subtypes) would increase the significance and support for the findings presented. Further data on the link (direct or indirect) between GATA6 and Wnt/BMP signalling would also add to the significance of this study. A number of textual changes/clarifications are also suggested to improve the manuscript.

We appreciate the feedback and provide responses for issues raised for markers, direct or indirect interactions, and textual changes/clarifications in the following sections. As indicated above, we did not find obvious alterations in cardiac subtypes, but since our study is focused on early progenitor specification, this is an interesting question that we think should be more rigorously evaluated in subsequent work.

Reviewer #1 (Recommendations for the authors):

Minor details:

(1) On p6 "Principal component analysis (PCA) showed that the cells derived from each genotype were well separated from each other (Supplemental Figure 2C)". All genotypes should be in one PCA plot to better evaluate the three genotypes.

We prepared the new plot as suggested, presented as new Supplemental Fig. 2C.

(2) p10: "Chia et al.22 and found a significantly decreased enrichment in GATA6^{-/-} cells relative to WT at day 2" decreased enrichment of what? Direct target genes?

Thank you for catching this. Yes, the text was changed to indicate a “decreased enrichment in GATA6^{-/-} cells relative to WT at day 2 for putative direct GATA6 target genes.”

Reviewer #2 (Recommendations for the authors):

Overall, this is an interesting study that addresses the early developmental roles of GATA6 on cardiac differentiation. While the identification of Wnt and BMP pathway genes to be involved in GATA6 regulation is not entirely unexpected, the authors do bring forth some useful knowledge that helps to further elucidate the mechanism of pre-cardiac mesoderm regulation. Some suggestions for improvement are included below -

Major points:

(1) Since the loss of Gata6 in this study is global (either as heterozygous or homozygous, it is likely that the very early requirement of Gata6 (e.g. mesodermal stage of differentiation) is responsible for the cardiac transcriptional phenotype observed and not due to specific role of Gata6 in the cardiac lineage which would need to be addressed using conditional knock out of Gata6 in hPSC model. The authors should be more explicit when discussing the results as disruption of mesodermal differentiation leading to loss of downstream cardiac lineage cells. For example, I would change the title "GATA6 loss-of-function impairs CM differentiation" to "GATA6 loss-of-function impairs mesodermal (or mesodermal lineage) differentiation" and show the changes in cardiac progenitor cells genes (Isl1, Tbx1, Hand1, and BAF50c/Smarcd3) in addition to cardiomyocyte genes but no change in mesodermal (e.g. Brachyury, T, Eomes, Mesp1/2, etc) genes.

We agree with the reviewer’s interpretation. The title for the section was changed as suggested. In Fig. 1, we show changes in cardiac progenitor cell genes (Isl1, Hand1, and BAF50c/Smarcd3) while not seeing changes in mesodermal genes in Fig. 2 (e.g. Brachyury, Eomes, Mesp1/2). We note that the defect may be specific to cardiac (or anterior lateral) mesoderm, as the ability to express paraxial mesoderm markers was not impaired.

(2) The use of NKX2.5, TBX5, TBX20, and GATA4 as markers for CPC is not ideal. These markers are also expressed in differentiated cardiomyocytes. ISL1 or TBX1 for second heart field progenitors and HAND1 or BAF60c/Smarcd3 for first heart field progenitors would be ideal.

As suggested, we included additional day 6 qPCR panel (new Fig. 1E) to evaluate the heart field progenitor markers.

(3) Much of the findings described in this study have been known in the field including the requirement of Wnt and BMP to induce mesodermal and subsequently cardiomyocyte differentiation. The key new information here is that Gata6 knockout disrupts Wnt and BMP signaling. It would help to further validate experimentally some of the Wnt and BMP genes as either direct or indirect targets of Gata6 using reporter assays.

While reporter assays are feasible and do provide relevant outputs, we feel that the use of any one or even several response elements in a reporter assay adds relatively little value compared to comprehensive analysis of bona fide network components. To address the reviewers concern we have included profiling heat maps for WNT and BMP pathway components to more rigorously and specifically evaluate the disruption in the signaling networks caused by loss of GATA6. Proving direct targets of endogenous genes is challenging, but we mapped many binding peaks for GATA6 to putative enhancers of WNT/BMP pathway genes (based on histone marks). We provide a list of these genes (new Fig. 4F) and distinguish

these from WNT/BMP pathway genes that were not bound by GATA6 yet are down-regulated in the GATA6 mutant cells and are likely to be indirect targets (p. 12).

Minor points:

(1) Figures 1 and 2 - in the figure legend the labels w2, w4, m2, m5, m11, and m14 should be explained as the name of the clones of targeted hESC.

The legends were edited to provide this information.

(2) Supplemental Figure 3A - the resolution of the FACS plot is suboptimal.

We apologize and have corrected the plot resolution in the revised manuscript.

(3) Supplemental Table 1 - it's intriguing that amongst all the SWI/SNF factors, the one that is known to be cardiac-specific (SMARCD3) did not come up in the GATA6-RIME-enriched proteins. Is this a reflection of the early stage in which GATA6 plays a role in development (e.g. mesendoderm development but not precardiac mesoderm development when SMARCD3 is expressed)?

We agree and have noted this feature in the revised manuscript (p. 17). We note that SMARCD3 is expressed in the RNA-seq data as early as day 2. Although speculative, it may be that GATA6 primarily interacts with SWI/SNF complexes prior to the role for SMARCD3 in cardiac specification.

Reviewer #3 (Recommendations for the authors):

(1) Figures 3G and 3H, as well as others, have resolution issues. The gene names are unreadable, and higherresolution images should be provided.

We apologize for the resolution issues and these have been fixed in the revised version.

(2) In their early manipulation of the WNT and BMP pathways (Figure 6A), it is unclear whether the activin/BMP protocol shown in Figure 1A was used. If this is the case, the authors should compare their results to a wild-type + DOX EV condition for consistency.

We clarified in the revision (Fig. 6A) that all the experiments in Fig. 6 use the cytokine protocol. In the revised figure, we included the wild-type + DOX EV condition as suggested.

(3) In Figures 6C and 6D, the authors should include an analysis of a wild-type isogenic line under their new CHIR/LB condition for comparison.

As suggested, we included the WT isogenic line in the comparison. For Fig. 6C these are shown on a separate graph because the Y-axis values are very different. Note that the CHIR/LB treatments that improve mutant cell differentiation impact the WT cells in the opposite manner.

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